

MINISTRY OF EDUCATION AND TRAINING STATE BANK OF VIETNAM

HO CHI MINH UNIVERSITY OF BANKING



GRADUATION THESIS

**THE BIDIRECTIONAL RELATIONSHIP BETWEEN
BANKRUPTCY RISK AND PROFITABILITY OF
SECURITIES COMPANIES IN VIETNAM**

MAJOR: FINANCE – BANKING

CODE: 7340201

TRAN GIA HUNG

HO CHI MINH CITY – 2025

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MAJOR: FINANCE – BANKING

CODE: 7340201

Student: TRAN GIA HUNG

Student ID: 050609211973

Class: HQ9-GE21

GRADUATE INSTRUCTOR

PhD. NGUYEN DUY LINH

HO CHI MINH CITY – 2025

ABSTRACT

The thesis examines the bidirectional relationship between bankruptcy risk and the profitability of securities companies in Vietnam from 2011 to 2024. The data set used in the research is collected from the Vietnam Economic - Financial - Macroeconomic Database (Fiin Pro) and GSO. The model has two dependent variables, which are the bankruptcy risk and profitability of companies. For the bankruptcy model, the author uses independent variables including return on assets, company size, equity-to-debt ratio, GDP growth, asset utilization efficiency, and return-risk balancing capability. For the profitability model, the dependent variable of the model is the return on assets, the independent variables include the bankruptcy risk, company size, equity-to-debt ratio, GDP growth, asset utilization efficiency, and return-risk balancing capability. The thesis uses FEM, REM, SGMM methods to assess the impact of independent variables on dependent variables and examines the bidirectional relationship between bankruptcy risk and profitability of securities companies. The research results show that there is a trade-off relationship between the bankruptcy risk of companies and profitability.

Keywords: Bankruptcy risk, profitability, securities company, SGMM

Ho Chi Minh City, July , 2025

Student

Tran Gia Hung

DECLARATION

I - Tran Gia Hung

Student ID: 050609211973

Class: HQ9-GE21

As the author, I hereby declare that my graduation thesis on the topic **The Bidirectional Relationship Between Bankruptcy Risk and Profitability of Securities Companies in Vietnam** is entirely the result of the research process that the author has conducted in the past. The thesis was carried out with the companionship and enthusiastic guidance of PhD. Nguyen Duy Linh. This is also the result of more than 3 years of studying and accumulating knowledge of the author at the Banking University of Ho Chi Minh City.

The author's research work is honest and transparent, the theories, previous studies, other references, data and necessary information are all presented specifically by the author throughout the thesis as well as in the references and appendices.

I am fully responsible for my graduation thesis.

Ho Chi Minh City, July , 2025

Student

Tran Gia Hung

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After about 3 months of completing my graduation thesis,

I have completed my thesis with the topic **The Bidirectional Relationship Between Bankruptcy Risk and Profitability of Securities Companies in Vietnam**. Here, I would like to express my gratitude to all individuals and organizations that have supported and accompanied me throughout my time studying at Ho Chi Minh City University of Banking.

First of all, I would like to sincerely thank the teachers of Ho Chi Minh City University of Banking for their dedication in imparting valuable knowledge to me during my time studying, and thank the Finance Faculty for proving to me that I have chosen the right school and the right major.

I would like to express my deep gratitude to PhD. Nguyen Duy Linh, who has wholeheartedly supported and helped me a lot in the journey to complete this thesis, from the time of inspiration to the final step, all of which have helped me complete my thesis effectively. I appreciate the knowledge and things that you have given me, you have contributed a lot to creating the current me, I thank you very much.

This essay certainly has some shortcomings because my accumulated knowledge is still limited, so I hope to receive valuable comments and suggestions from teachers to help me supplement and improve my knowledge, do better in the next essays, and future work.

Finally, I would like to wish teachers and everyone good health, success in work and life!

Ho Chi Minh City, July , 2025

Student

Tran Gia Hung

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LIST OF ABBREVIATIONS

No.	Abbreviation	Definition
1	EBIT	Earning before Interest and Taxes
2	FEM	Fixed effect model
3	GSO	General Statistics Office
4	KSE	Korea Stock Exchange
5	KOSDAQ	Korean Securities Dealers Automated Quotations
7	M&A	Mergers and Acquisitions
8	OLS	Ordinary least squares
9	PhD	Doctor of Philosophy
10	REM	Random effect model
11	ROA	Return on Assets
12	ROE	Return on Equity
13	SGMM	System Generalized Method of Moments
14	VND	Vietnam Dong

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CHAPTER 1: INTRODUCTION

1.1. Research Rationale

Securities companies often play an important bridging role in the financial market, not only helping to channel capital from investors into the market, but also taking on the role of creating liquidity and providing diverse financial services. In 2024, the market capitalization of listed stocks reached VND 5.21 quadrillion, up 14.3% compared to 2023, while the average daily trading value was VND 18.685 trillion, up 22.4% from the previous year. By the end of 2024, the liquidity of the Vietnamese stock market ranked 3rd in Southeast Asia, after the Thai and Singapore markets. In addition, Vietnam has started implementing the KRX system since May 2025, and is on the list of FTSE's emerging market upgrade, creating conditions for many securities companies to develop significantly. However, high profits always come with risks, especially the risk of losses that rapidly erode equity value and reduce the ability to repay debt when the market fluctuates strongly.

In general, securities companies are one of the key links in the structure of the stock market. These companies not only attract and direct capital from outside into the investment channel, but also take on the role of creating liquidity and providing a variety of financial solutions. When a securities company falls into insolvency or bankruptcy, the spillover effects can quickly undermine market confidence, reduce overall liquidity, and trigger sharp fluctuations in stock prices.

There is often a reciprocal relationship between bankruptcy risk and profitability. Strict risk control helps minimize losses, preserve equity, thereby creating a stable foundation for sustainable profitability. At the same time, the pursuit of higher profits requires careful consideration of financial safety to avoid excessive risk-taking that could disrupt the balance between profit potential and risk tolerance. However, in research articles on corporate finance and risk management, there is often a tendency to approach in a univariate direction - that is, focusing only

on a specific aspect such as profitability or bankruptcy risk, and rarely exploiting the mutual relationship between these two factors in the same analytical model.

In developed financial markets, the relationship between risk and profit is often explained through the theory of trade-off between risk and expected return, agency cost theory, M&M theory, pecking order theory, etc. Optimizing profitability requires accepting a certain level of risk, but when risk exceeds the control threshold, it can turn back to erode profitability and threaten the existence of the business itself. Theories such as the Agency cost theory provide a clearer view of the relationship between bankruptcy risk and profitability. A company with a high risk of bankruptcy will often be able to create greater pressure on members of the company's executive board, forcing them to act cautiously and appropriately to avoid bankruptcy. On the contrary, according to Pecking order theory, to have higher profitability, many businesses have to expand, and this process is often accompanied by more debt, thereby increasing the risk of bankruptcy. Because of this bidirectional relationship, the use of traditional research models such as FEM or REM may not be sufficient to clarify the interaction mechanism. Instead, dynamic models such as SGMM (System Generalized Methods of Moment) are more suitable to test for endogeneity and measure the mutual relationship between two variables.

In addition, previous studies have often focused on manufacturing, banking or non-financial enterprises in general – sectors with complete data and long time series – while securities companies have rarely been systematically surveyed. Especially in Vietnam, there have been only a few individual studies that have addressed the risk of bankruptcy in the securities industry from the Z-score perspective, and the works assessing profitability often stop at the analysis of ROE or ROA alone, lacking integration in the context of the overall risk of the enterprise. Analyzing the link between these two factors at the same time will help fill the

academic gap and systematically contribute to the theoretical foundation of financial management in the securities industry in developing markets.

This study not only carries academic value but also offers direct managerial implications for the leadership of securities companies, those responsible for capital allocation policies, financial leverage management, and bankruptcy risk control. A clear understanding of the relationship between profitability and risk enables executives to optimize capital structure so that it remains sufficiently flexible yet not excessively risky, allocate resources efficiently across business segments, establish early-warning systems, and quantify acceptable risk levels for each target profit margin, rather than merely setting revenue or profit targets without assessing the firm's risk tolerance.

From the perspective of investors, especially institutional investors, knowing which securities companies are maintaining a reasonable balance between profitability and capital safety is also an important basis for making investment decisions or choosing partners. In the context of foreign capital flows tending to return to Vietnam and securities companies in the process of increasing capital, listing or issuing bonds, studies like this will be a useful tool to assess financial health comprehensively, beyond superficial financial figures.

Therefore, identifying the factors affecting the relationship between bankruptcy risk and profitability is a prerequisite. Therefore, the study on **The bidirectional relationship between bankruptcy risk and profitability of securities companies in Vietnam** can provide valuable applied analysis, helping managers make appropriate decisions and strategies.

1.2. Research objectives

1.2.1. General Objective

This research aims to analyze the bidirectional relationship between bankruptcy risk and profitability of securities companies in Vietnam.

1.2.2. Specific Objectives

To achieve the above general objective, the research focuses on three specific goals:

Objective 1: Identify the key factors influencing both the bankruptcy risk and profitability of securities companies in Vietnam.

Objective 2: Examine the bidirectional relationship between bankruptcy risk and profitability among these firms.

Objective 3: Propose appropriate financial strategies to mitigate bankruptcy risk and enhance the profitability of securities companies.

1.3. Research questions

Question 1: What are the key factors affecting the bankruptcy risk and profitability of securities companies in Vietnam?

Question 2: Is there a bidirectional relationship between bankruptcy risk and profitability, and how strong is this relationship?

Question 3: What financial strategies and policy recommendations can improve performance and reduce financial distress in securities firms?

1.4. Research subject and scope

1.4.1. Research Subject

The research focuses on the relationship between bankruptcy risk and the profitability of securities companies in Vietnam.

1.4.2. Research Scope

Spatial Scope: The research spatial frame for this study is the financial data of securities companies operating in the Vietnamese market during the period from 2011 to 2024. These securities companies are large-scale securities companies, accounting for the majority of market share in Vietnam.

Time Scope: The research time frame of the paper is the period from 2011 to 2024. This period marks important events for the economy and stock market in

Vietnam. Major events such as the implementation of the US-China trade war in 2017, the Securities Law 2019, the Enterprise Law 2020, the Covid 19 pandemic, the bond crisis in 2022, ... Therefore, the data collected during this period will show the impacts of macro and micro fluctuations on securities companies.

1.5. Research methodology

1.5.1. Data Collection Method

The dataset of this study is collected from 22 securities companies operating in the Vietnamese market and other macro data in the period from 2011 to 2024. This dataset is based on financial reports, data from Finpro and statistical results from GSO.

1.5.2. Research Methodology

To preprocess the data, the author uses Microsoft Excel software. To perform the econometric steps and analyze panel data, the author uses Stata 17 software, including the following steps: Descriptive statistics of variables; Correlation analysis; Model estimation through three models including: FEM model, and REM model; Conduct tests to select appropriate models and check for defects of the selected model; Correct defects and errors of the selected model (if any) using main methods such as SGMM suitable for the detected defects. Specifically, in this essay, the method chosen to correct the defects is S-GMM.

1.6. Research contributions

Academically, this research contributes to expanding the theoretical foundation on the bidirectional relationship between bankruptcy risk and profitability, a topic that has received relatively little in-depth attention in the context of financial intermediaries in emerging markets such as Vietnam. By systematically reviewing existing theoretical frameworks and prior empirical studies, combined with quantitative analysis of real-world data, the research provides clear empirical evidence of the mutual interaction between these two critical financial variables. This, in turn, offers a novel perspective for financial

management theories in the securities sector. Moreover, the thesis tests, evaluates, and proposes a suitable quantitative model for measuring bankruptcy risk under the specific conditions of the Vietnamese financial market, thereby contributing to the enhancement of financial analysis tools for future academic research.

The study provides a comprehensive assessment of the relationship between bankruptcy risk and profitability of securities companies operating in Vietnam. By identifying key financial linkages and factors affecting business performance, the findings can assist corporate executives in making strategic decisions, optimizing capital structure, managing risk and improving profitability.

1.7. Structure of the study

This study is divided into 5 chapters:

Chapter 1: Introduction

Chapter 1 focuses on introducing the thesis topic, presenting the research rationale for the topic, indicating the thesis objectives, research questions, research objects and scope, research methods, giving the expected contributions of the thesis and introducing an overview of the thesis structure.

Chapter 2: Literature review

Theoretical framework of the bidirectional relationship between bankruptcy risk and profitability at securities companies in Vietnam. Presenting theories and related studies that have been conducted to construct models and research methods in order.

Chapter 3: Methodology

Chapter 3 focuses on selecting the research model, explaining the variables included in the model and presenting the methods of data collection, measurement, and processing, while introducing the process of implementing the econometric regression of the thesis.

Chapter 4: Findings and Discussion

Based on the above chapters, in chapter 4, the author will focus on performing econometric regression for the thesis, performing data analysis, selecting models, running the model and analyzing the results after running the relationship between bankruptcy risk and profitability.

Chapter 5: Conclusion

Based on chapter 4, the author will propose some recommendations on the relationship between bankruptcy risk and profitability of securities companies. In addition, in this chapter, the author also reviews the research and points out the limitations in the research process that the author has concluded, and at the same time develops the next research direction in the future.

CHAPTER 1 SUMMARY

Chapter 1 has summarized the issues related to the thesis topic including the urgency of the topic, research objectives, research questions, research objects and scope, research methods, proposed contributions of the thesis and introduced the structure of the entire thesis. Currently, the situation of the relationship between bankruptcy risk and profitability at Vietnamese securities companies has not been studied in depth, the author hopes that this thesis will contribute to explaining this relationship and provide practical solutions and recommendations to help corporate administrators make more appropriate decisions.

In the next chapter, the author will present the theoretical basis of bankruptcy risk, profitability, micro and macro factors affecting the relationship between bankruptcy risk and profitability through theories and previous studies that the author has reviewed, thereby giving a general assessment and forming a research model in chapter 3.

CHAPTER 2: THEORETICAL FRAMEWORK

Chapter 2, in order to provide theories related to bankruptcy risk and profitability, and at the same time provide a theoretical framework for building a research model in Chapter 3, the author presents the concept of profitability, the concept of bankruptcy risk. At the same time, it presents a number of theories and hypotheses about factors affecting profitability and bankruptcy risk, and summarizes the results of previous studies related to the relationship between bankruptcy risk and profitability in the world and in Vietnam, thereby providing a basis for pointing out the research gaps of these papers, and finally synthesizing the theory of factors affecting the relationship between risk and profitability that the author has researched in Chapter 3.

2.1 Theoretical framework

2.1.1 Bankruptcy risk

2.1.1.1. Concepts

Although there are many studies conducted in the world, the definition of bankruptcy risk has not been unified by scholars. In general, bankruptcy risk is often defined as a decline in profitability, leading to bankruptcy due to financial pressures.

Regarding the concept of bankruptcy risk, Samad and Armstrong (2022) said that bankruptcy risk is the decline of enterprises due to inefficient operations and waste of resources. Nguyen Thanh Duong (2013), when studying banks, pointed out that the risk of bankruptcy comes from financial exhaustion due to reduced income. Shaffer (2012) pointed out that bankruptcy risk can be controlled and limited if capital intensity, financial leverage and profit margin are controlled, which will help limit the bankruptcy risk of businesses.

2.1.1.2. Measurement Methods

Beaver (1966) believed that bankruptcy risk can be identified and evaluated using financial indicators. Following the research from Beaver (1966), Altman

(1968) developed the Z-score formula to identify and evaluate the bankruptcy risk of enterprises.

Z-score is not only a popular credit analysis tool in banking, but also widely used in investment decisions, credit rating assessments, M&A strategy planning or internal audits. In the current volatile global economic context, the Z-score index is even more valuable in detecting early signs of financial weakness, allowing businesses to promptly deploy appropriate response measures. According to Altman (1968), the Z-score formula is as follows:

$$\text{Z-score}_{it} = 1.2 X_1 + 1.4 X_2 + 3.3 X_3 + 0.6 X_4 + 1.0 X_5$$

Where:

$$X_1 = \frac{\text{Current Assets}_{it} - \text{Current Liabilities}_{it}}{\text{Total Assets}_{it}}$$

$$X_2 = \frac{\text{Retained Earnings}_{it}}{\text{Total Assets}_{it}}$$

$$X_3 = \frac{\text{EBIT}_{it}}{\text{Total Assets}_{it}}$$

$$X_4 = \frac{\text{Market Capitalization}_{it}}{\text{Total Liabilities}_{it}}$$

$$X_5 = \frac{\text{Revenue}_{it}}{\text{Total Assets}_{it}}$$

A higher Z-score indicates that the enterprise has low bankruptcy risk, financial stability, a robust risk management system, and therefore exhibits high resilience to risks and market shocks, especially those from the financial market. On the contrary, a low Z-score indicates that the enterprise is under pressure in terms of solvency, or may be a sign of impending bankruptcy. This shows that the enterprise needs to improve liquidity, improve the risk management system, or restructure if necessary.

Altman's original (1968) Z-score was used primarily for pragmatic reasons: its component accounting ratios were consistently available across the sample period and facilitated comparison with results from the relevant literature. However, the 1968 specification was developed for manufacturing firms and does not capture features specific to banks and other financial institutions (for example, regulatory capital structure and liquidity dynamics). Consequently, its absolute values and the ranking of financial distress may be biased when applied to firms in the financial sector.

2.1.2. Profitability

2.1.2.1. Concepts

Profitability is one of the many important indicators used to evaluate and analyze a company's performance. Profitability represents the ability to generate profits from the company's main business activities.

There have been numerous research projects conducted worldwide; however, each scholar's perspective on profitability differs. Some opinions argue that profitability is the ability to generate income from available resources for expansion and growth within a fixed period of time (Heibati et al. 2009; Duy Khang Le et al. 2022).

Pandey (1980) believes that profitability is the ability of a business to generate profits from business activities within a certain period. However, this concept does not stop at determining pure profit, but also shows the relationship between profit and other factors such as revenue, shareholder investment capital, or total assets used in the production-business process.

Malik (2011) asserts that profitability is one of the key goals in financial management and is a core factor reflecting the performance of the business. In the context of the increasingly important goal of maximizing shareholder value, profitability plays a role as an indicator to evaluate the level of success in using financial resources to create added value.

2.1.2.2. Measurement Methods

Although many metrics can be used to measure the profitability of securities companies, including Return on Assets, Return on Equity, Return on Invested Capital, etc. However, ROA is generally preferred as an indicator to measure the profitability of companies, as noted by Delen et al. (2013), Cao Minh Tien (2023), Mehzabin et al. (2022), Onaolapo and Kajola (2010).

The author chose to use ROA for many reasons. Firstly, ROA is a metric that is less affected by financial leverage, so it avoids the times when businesses "beautify" their business results by increasing leverage. For that reason, many related studies in the field of banking and finance often use ROA because it allows measuring the company's operating efficiency on all existing assets with a comprehensive, general perspective. And because it is a "truth-telling" metric, the effect of ROA is also a popular sign to identify suitable business models, the effectiveness of business strategies or operational efficiency, according to Higgins (2012). The author chose the formula created by Ross et al. (2022):

$$ROA_{it} = \frac{Net\ income_{it}}{Total\ Assets_{it}}$$

ROA_{it} is the return on total assets of bank i in year t

$Net\ income_{it}$ is the net income of company i in year t

$Total\ Assets_{it}$ is the total assets of company i in year t

2.1.3. Theories on the relationship between bankruptcy risk and profitability of securities companies

2.1.3.1. Agency cost theory

Jensen and Meckling (1976) developed Agency cost theory in their research to build a new theoretical framework to explain the relationship between capital structure and profitability of companies in the market. From this perspective, it is also possible to gain a deeper understanding of the relationship between bankruptcy risk and a company's profitability. According to Agency cost theory, within a

company, there will be conflicts of interest between shareholders and managers. When ownership and control are separated, managers may make decisions that are not optimal for shareholders' interests, such as wasteful spending, investing in inefficient projects, or pursuing personal interests instead of increasing the value of the company.

Agency cost theory suggests that through capital structure such as financial leverage, the agency costs of the company can be reduced. By borrowing more debt, which increases the risk of bankruptcy of the company, managers of the company will be under more financial pressure and supervision from creditors. A higher bankruptcy risk will create pressure can forces these managers to act cautiously with business plans, investment activities, and less wasteful spending... to ensure the ability to repay debts and avoid bankruptcy. Therefore, increasing financial leverage or increasing bankruptcy risk can enhance firm performance by constraining and monitoring the actions and plans of corporate managers (Grossman and Hart, 1982; Jensen, 1986; Williams, 1987).

However, Jensen and Meckling (1976) also pointed out that the relationship between financial leverage and performance or agency costs is not linear. When leverage is low, borrowing creates incentives for managers to act more prudently and helps reduce agency costs. However, as leverage increases to a threshold where the risk of bankruptcy and the likelihood of financial distress become higher, the agency costs of external debt will dominate over the agency costs of external equity, causing further increases in leverage to result in increased total agency costs.

2.1.3.2. Pecking order theory

Proposed by Myers and Majluf (1984), the Pecking order theory serves as an important theoretical framework to explain a firm's financing decisions, thereby affecting its capital structure and bankruptcy risk. Pecking order theory assumes that there is information asymmetry between the firm's managers and investors. Asymmetric information affects the choice between internal and external financing

and between new issues of debt and equity securities. According to the Pecking order theory, a firm will usually prioritize the choice of capital sources to finance its production and business activities from internal capital first, then debt, and finally equity (Brealey, 2020)

Pecking order theory assumes internal capital is often preferred because the cost incurred by the firm is low and does not show negative signals like the use of external sources of finance. The issuance of debt by a company shows that the company's managers are confident in the company's profitability, and the abandonment of the option of issuing shares to finance shows that the company is undervalued. Compared to issuing shares, the company is showing a negative signal when it believes that the current share price is overvalued or that the company's management is not really confident about the project.

Overall, the Pecking order theory suggests that when a company has high profitability, good operating efficiency, and abundant cash flow, it will have less need for new debt. Most of the company's capital at this stage will be financed mainly by internal capital. Enterprises in a healthy financial state will prioritize debt repayment and capital accumulation instead of seeking external sources of funding. Less debt will make the company's financial leverage more balanced, and the risk of bankruptcy will also be significantly lower. Conversely, when profitability declines, insufficient internal capital forces businesses to borrow from outside to finance their operations, leading to increased bankruptcy risks.

2.2 Literature review

2.2.1. International studies

Xu (2015) conducted a study on small and medium-sized enterprises and large private firms in Japan during the period from 2002 to 2012 and found that bankruptcy risk was positively correlated with the profitability of companies. The results from the data set showed that the risk tolerance of enterprises was positively correlated with profitability. Similar to Xu (2015), Kim (2011) conducted a study

on non-financial companies operating on the Korean stock market such as KSE, KOSDAQ also found a positive correlation between bankruptcy risk and profitability. With the results from the 2SLS model, Kim's (2011) study encouraged companies to accept higher risks in exchange for increased profitability.

Mehzabin et al. (2022) in their study of 492 banks in 28 Asian countries over a period of 15 years, from 2004 to 2018, found evidence of a correlation between bankruptcy risk and bank profitability. The study measured profitability using ROA and ROE, and found evidence of a positive impact of financial leverage and company size on profitability. The results confirm that an expansion in debt position is accompanied by an increase in profit; the larger the overall debt, the more profitable the bank can be in countries of Asia. In fact, the findings are in line with the agency cost theory and contradict the pecking order theory that suggests prioritizing retained earnings to finance the company assets.

Berger and Udell (2005) conducted a quantitative study to test the Agency cost theory on a sample of commercial banks in the United States from 1990 to 1995 and found a positive relationship between bankruptcy risk and profitability. Specifically, higher leverage or a lower equity capital ratio is associated with higher profit efficiency over almost the entire range of the observed data. Berger and Udell (2005) imply that banks in the United States may use debt to create incentives for bank management to act prudently and appropriately, improving profitability in response to the increased bankruptcy risk from debt.

Ayalew (2021) studied 16 Ethiopian private commercial banks during 2013/14–2018/19 using a fixed-effects model, showing that total debt ratio, short-term debt, loan-to-deposit ratio, and credit risk have a positive impact on bank profitability, measured by ROA, ROE, and NIMA, while size has a negative impact and bank age has a positive impact. The study recommends prudent liquidity and credit risk management, but is limited in that it does not consider macroeconomic factors, is non-quantitative, and uses book value instead of market value.

Yinusa et al. (2019) in their study of non-financial listed companies in Nigeria also found a positive correlation between bankruptcy risk and profitability. Using a GMM model of 115 Nigerian listed companies from 1998 to 2015, Yinusa et al. (2019) found that high debt ratios serve as a disciplinary device which may help enhance profitability due to the debt repayment obligation which makes managers strive to ensure they generate sufficient cash flow that can prevent bankruptcy. However, Yinusa et al. (2019) also found that excess short-term debt may be employed for suboptimal investment which increases the default risk which may make debt repayment very difficult and eventually can result in debt overhang problem which may be inimical to the firm.

When studying the factors affecting the profitability of companies operating in Türkiye from 2005 to 2011 using a decision tree model, Delen et al. (2013) found that financial leverage and debt ratios were found to impact profitability. Debt is a source of financing, other than equity. If a firm invests funds obtained through debt in profitable operations, it will in all likelihood have a higher profitability (measured by ROA and ROE).

Akhtar et al. (2021) in their study of 424 non-financial companies listed on the Pakistan Stock Exchange during the period from 2001 to 2017 showed a positive correlation between bankruptcy risk and profitability. By trading off higher bankruptcy risk through increased financial leverage, the profitability of the companies can be improved accordingly. The results also confirm an inverted U-shaped relationship between debt and profitability, indicating that an increase in the debt of Pakistani listed companies increases their profitability up to a certain level, and then, a further increase in debt can reduce profitability.

2.2.2. Domestic studies

Tran Van Hai et al. (2021) studied the relationship between bankruptcy risk and profitability of 49 securities companies in Vietnam during the period from 2015 to 2019. By studying the impact of the Z-score coefficient (component is 5 financial

ratios X1; X2; X3; X4; X5) on the profitability of securities companies, through the Pooled Ordinary Least Square regression model. Specifically, Tran Van Hai et al. (2021) showed that factors such as retained earnings on total assets and net revenue on total have a positive impact on the profitability of securities companies.

Cao Minh Tien (2023) conducted a study to examine the relationship between capital structure and performance of securities companies in Vietnam. Inheriting previous studies on the relationship between capital structure and performance, the author argues, tests and determines the optimal capital structure. By combining qualitative (synthesis, descriptive statistics, induction and interpretation) and quantitative (linear regression) research methods, the study analyzed data from 105 brokerage securities companies, corresponding to 1,314 observations in the period 2003–2021. The results showed that there are two factors of capital structure that significantly affect performance, including financial leverage and company size.

Tu D. Q. Le and Dat T. Nguyen (2020), when studying the Vietnamese banking system using the Quantile Regression method, showed that bank capitalization has a positive impact on profitability, and there is an inverted U-shaped relationship between capitalization and ROE, but it is only statistically significant at the 90th percentile. The study used a dataset of 30 Vietnamese banks in the period 2007–2019, thereby showing that increasing capital improves profits up to a limit point, after which the impact will reverse and reduce profits. This result suggests that Vietnamese banks need to be cautious when raising capital ratios according to Basel standards, and should do it step by step to avoid exceeding the optimal threshold.

Nguyen Thuy Anh and Tran Thi Phuong Thao (2019) showed the relationship between bankruptcy risk and profitability among companies in Vietnam. With a dataset of 446 non-financial companies listed on the stock market in Vietnam during the period from 2011 to 2017, Nguyen Thuy Anh and Tran Thi

Phuong Thao (2019) showed the positive impact of bankruptcy risk on improving profitability. Specifically, financial leverage has a positive relationship with the profitability of enterprises. In which, more debt makes the company face financial pressure and higher bankruptcy risk, thus forcing managers to operate better to cope with financial pressure and higher bankruptcy risk.

Huong Thi Thanh Co et al. (2021) in their study of a number of companies operating in the Vietnamese stock market during the period from 2015 to 2019 also showed similar implied conclusions about the relationship between bankruptcy risk and profitability. Specifically, the regression results using OLS, FEM, and REM showed that although debt has a positive impact on the company's profitability, there is still a difference between long-term debt and short-term debt. Short-term debt has a positive relationship with profitability, while long-term debt shows an inverse relationship.

Hung The Dinh and Cuong Duc Pham (2020) in their study of pharmaceutical companies in Vietnam during the period from 2015 to 2019 showed evidence of bankruptcy risk and profitability. With the regression results from OLS, Hung The Dinh and Cuong Duc Pham (2020) showed a positive impact of borrowing and expanding the company's size on profitability. Therefore, Hung The Dinh and Cuong Duc Pham (2020) recommend that companies should increase financial leverage by increasing debt and expanding the company, although it will increase debt repayment pressure and bankruptcy risk, but can improve profitability.

Doan Thanh Ha (2015) when studying 150 companies listed on the Vietnamese stock market in the period from 2008 to 2012 showed that the relationship between bankruptcy risk and profitability of the company can be explained through size and financial leverage. The research results show that companies can increase debt, especially short-term debt and expand the scale to help companies have a positive impact on the profitability of companies in this period.

The research results of Le Thai Phong (2016) also imply a two-way relationship between bankruptcy risk and profitability. In his study, with a data set of 45 enterprises operating in the construction industry listed on the Vietnamese stock market in the period from 2007 to 2011 processed by the GLS method, it showed a positive impact of the debt-to-equity ratio on the profitability of enterprises.

Table 2.1. Summary of related studies

Author	Research issue	Period	Methodology	Findings
Xu (2015)	Risk taking and firm growth	2002 – 2012	OLS	Risk-taking has positive effects on profitability
Kim (2011)	Do Foreign Investors Encourage Value-Enhancing Corporate Risk Taking?	2001 – 2007	2SLS	Risk-taking positively associated with profitability
Mehzazbin et al. (2022)	The effect of capital structure, operating efficiency and non-interest income on bank profitability: new evidence from Asia	2004 – 2018	Fixed effect regression model	Financial leverage and company size can have positive effect on firm profitability
Berger and Di Patti (2005)	Capital structure and firm performance: A new approach to testing agency theory and an application to the banking industry	1990 – 1995	Two-Stage Least Squares (2SLS)	Financial leverage improves companies profitability

Author	Research issue	Period	Methodology	Findings
Ayalew (2021)	Capital structure and profitability: Panel dataevidence of private banks in Ethiopia	2013/14 – 2018/19	OLS	Debt ratio has a positive impact on profitability, but bank size has a negative impact
Yinusa et al. (2019)	Capital Structure and Firm Performance in Nigeria	1998 – 2015	Two-step generalized method of moments (GMM)	Financial leverage has a positive effect on profitability
Delen et al. (2013)	Measuring firm performance using financial ratios: A decision tree approach	2005 – 2011	Decision Tree analysis	Financial leverage positively impact firm profitability
Akhtar et al. (2021)	Impact of financial leverage on sustainable growth, market performance, and profitability	2001 – 2017	GMM	There is a U-shaped relationship between financial leverage and profitability
Tran Van Hai et al. (2021)	Balancing Risk and Profitability of Securities Companies – A Case Study in Vietnam	2015 – 2019	OLS	The research results show the relationship between two risks of bankruptcy and profitability

Author	Research issue	Period	Methodology	Findings
Cao Minh Tien (2023)	The relationship between capital structure and performance of securities brokerage firms – a case study in Vietnam	2003 – 2021	OLS, FEM, REM, and GLS	Financial leverage and firm size significantly affect performance
Tu D. Q. Le and Dat T. Nguyen (2020)	Capital Structure and Bank Profitability in Vietnam: A Quantile Regression Approach	2007 – 2019	Quantile regression approach	The inverted U-shaped relationship between capitalization and profitability
Huong Thi Thanh Co et al. (2021)	The Impact of Capital Structure on Firm's Profitability: A Case Study of the Rubber Industry in Vietnam	2015 – 2019	Panel data regression model: Pool OLS, FEM, REM	The use of short-term debt has a positive impact on profitability
Nguyen Thuy Anh and Tran Thi Phuong Thao (2019)	The Impact of Capital Structure on Firm Performance of Vietnamese Non-financial Listed Companies Based on Agency Cost Theory	2011 – 2017	Pooled OLS, REM, FEM and 2SLS	Financial leverage has a positive effect on profitability
Hung The Dinh and Cuong Duc Pham (2020)	The Effect of Capital Structure on Financial Performance of Vietnamese Listing Pharmaceutical Enterprises	2015 – 2019	OLS	Financial leverage and company size can have positive effect on firm profitability

Author	Research issue	Period	Methodology	Findings
Doan Thanh Ha (2015)	The effect of capital structure on corporate performance: Evidence in Vietnam	2008 – 2012	FEM, REM	Profitability positively in related with corporate performance in debt to assets
Le Thai Phong (2016)	The impact of financial leverage on firm performance: A case of construction firms in Vietnam	2007 – 2011	GLS	Financial leverage and company size can have positive effect on firm profitability

2.3. Research gap

The author finds that studies on the bidirectional relationship between bankruptcy risk and profitability are still limited. Previous studies often only evaluate this relationship mainly through financial leverage, not expanding to other factors such as size, asset utilization efficiency, etc. In addition, there are still quite a few studies conducted in Vietnam, especially with the subject being securities companies. Besides, previous studies have not updated the business situation and economic context of recent years.

CHAPTER 2 SUMMARY

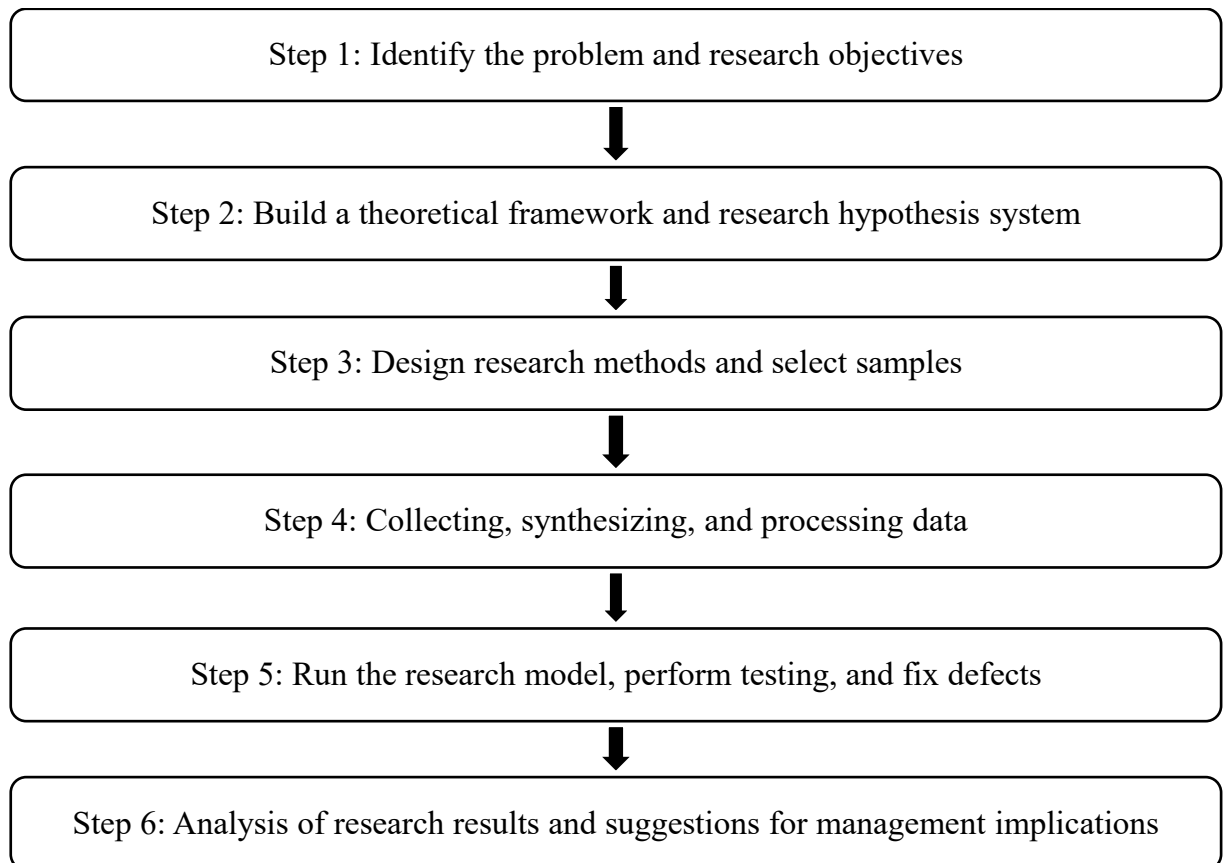
Chapter 2 provides a theoretical foundation on bankruptcy risk, profitability and factors affecting bankruptcy risk and profitability for securities companies in Vietnam through relevant theories and studies. At the same time, the author also points out the research gap and shapes the approach of the thesis model to the bidirectional relationship between bankruptcy risk and profitability. Following chapter 2, in chapter 3 the author will present the research model in detail, based on the inheritance of previous relevant studies.

CHAPTER 3: METHODOLOGY

3.1. Research progress

The research process of the topic The Bidirectional Relationship Between Bankruptcy Risk and Profitability of Securities Companies in Vietnam, includes 6 steps, specifically:

Image 3.1. Research progress



Source: compiled by author

Step 1: Identify the problem and research objectives

Understand and identify the problem to be researched, including the theoretical basis, research basis, and current status of the problem being researched. Then, identify the research objectives of the topic.

Step 2: Build a theoretical framework and research hypothesis system

After identifying the problem to be researched, proceed to build a theoretical framework and research hypotheses based on research and theories related to the

research. This step requires identifying the factors that have a significant impact on the research, as well as the formula and measurement method of each factor.

Step 3: Design research methods and select samples

Based on the results obtained from step two, proceed to build a research method and select a sample set for the model. During this process, it is necessary to determine the research method of the model, the types of tests used, and the defect handling model must be suitable for the data and research topic. In addition, the selection must be made on the basis of rationality, feasibility, and suitability for the research topic.

Step 4: Collecting, synthesizing, and processing data

After determining the data to be collected, collect data from reputable sources such as financial reports published on the company's website, FiinPro, or from the State Securities Commission... then calculate the values of the variables, compare the data, and proceed to process if the data has problems.

Step 5: Run the research model, perform testing, and fix defects

After obtaining suitable data, run the regression model (FEM and REM) using Stata17 software to assess the impact of factors on the dependent variable. At the same time, relevant tests must be performed to evaluate the applicability of the model. Then, if the model records defects, it is necessary to run the SGMM model to overcome the defects.

Step 6: Analysis of research results and suggestions for management implications

After having the results from SGMM, proceed to analyze and evaluate the fluctuations of the variables for the dependent variable and draw conclusions about the initial research hypotheses. This evaluation process must be carried out objectively, the analysis must be detailed and closely linked to practice. In addition, the author gives some suggestions for management implications to help securities companies improve their operational efficiency based on the research.

3.2. Proposed research model

The bidirectional relationship between bankruptcy risk and profitability is a new topic, and there are currently few studies evaluating the relationship. In previous studies, most authors mainly studied one-way, on the impact of bankruptcy risk on profitability, or how profitability factors affect bankruptcy risk. In this study, the author uses the concept of Z-score ... to evaluate the bankruptcy risk of companies - in this case, securities companies. For profitability, the author uses the concept of ROA to measure the ability to generate profits of these companies.

For the bankruptcy model, the author relies on a number of typical studies such as Altman (1968), Tran Van Hai et al. (2021), Co Huong Thi Hanh et al. (2021), Tran Van Hai (2022). For the profitability model, the author relies on many typical studies, such as Tran Van Hai et al. (2021), Nguyen Van Hau et al. (2021), Onaolapo and Kajola (2010), Pouraghajan and Malekian (2012), Rahman et al. (2023). Therefore, the author proposes the following research model:

3.2.1. Model for evaluating factors affecting bankruptcy risk

Model 1: $ZSCORE_{it} = \beta_0 + \beta_1 ROA_{it} + \beta_2 SIZE_{it} + \beta_3 EDR_{it} + \beta_4 AT_{it} + \beta_5 RMC_{it} + \beta_6 GDP_t + \varepsilon_{it}$

Where:

Dependent variables:

- $ZSCORE_{it}$: Bankruptcy risk of securities company i, year t

Independent variables:

- ROA_{it} : Return on Assets of securities company i, year t
- $SIZE_{it}$: Size of securities company i, year t
- EDR_{it} : Ratio of equity to total liabilities of securities company i, year t
- AT_{it} : Asset utilization efficiency of securities company i, year t
- RMC_{it} : Return–risk balancing capability of securities company i, year t
- GDP_t : GDP growth in year t

β_0 : Intercept coefficient

ε_{it} : Model residuals

$\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6$: Coefficients showing the level of influence of each independent variable on the dependent variable

3.2.2. Model to evaluate factors affecting profitability

Model 2: $ROA_{it} = \alpha_0 + \alpha_1 ZSCORE_{it} + \alpha_2 SIZE_{it} + \alpha_3 EDR_{it} + \alpha_4 AT_{it} + \alpha_5 RMC_{it} + \alpha_6 GDP_t + \mu_{it}$

Where:

Dependent variables:

- ROA_{it} : Return on Assets of securities company i, year t

Independent variables:

- $ZSCORE_{it}$: Bankruptcy risk of securities company i, year t
- $SIZE_{it}$: Size of securities company i, year t
- EDR_{it} : Ratio of equity to total liabilities of securities company i, year t
- AT_{it} : Asset utilization efficiency of securities company i, year t
- RMC_{it} : Return–risk balancing capability of securities company i, year t
- GDP_t : GDP growth in year t

α_0 : Intercept coefficient

ε_{it} : Model residuals

$\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$: Coefficients showing the level of influence of each independent variable on the dependent variable

3.3. Explanations of variables in the model and research hypotheses

Company Size (SIZE):

Company size plays an important role when considering the profitability and bankruptcy risk of securities companies. Large companies often have many competitive advantages such as a large customer base, access to cheap capital and better management systems than smaller companies. However, being too large also leads to many significant impacts. For securities companies, expanding their operations inherently increases pressure on capital raising and the management of cash flows.

Cao Minh Tien (2023), Huong Thi Thanh Co et al. (2021) and Thi Anh Nguyet Nguyen and Chien Van Nguyen (2025) have shown a significant impact of firm size on profitability. However, being too large can increase the risk of bankruptcy, according to Alfaro et al. (2019), Thi Thanh Nguyen and Vu Duc Kien (2022), Tram Thi Xuan Huong and Nguyen Tu Nhu (2020). The size of securities companies is determined by the following formula:

$$\text{SIZE} = \ln(\text{Total Assets}_{it})$$

The input data to calculate the value of SIZE is taken from the published financial reports of securities companies provided by Fiinpro.

Equity-to-Debt ratio (EDR):

Companies in the financial sector such as banks and securities companies often use high financial leverage to optimize profits. However, unlike commercial banks, the operations of securities companies are riskier when accompanied by proprietary trading and margin lending activities. Therefore, each company must maintain a large source of safety capital to create a "capital buffer" sufficient to absorb macroeconomic shocks or business problems.

Referring to Altman (1968), Yinusa et al. (2019), Berger and Di Patti (2005), Nguyen Thuy Anh and Tran Thi Phuong Thao (2019), the author uses the ratio between equity and debt variables as a factor to explain the impact of financial leverage on the relationship between bankruptcy risk and profitability of securities companies. The EDR variable formula is as follows:

$$\text{EDR} = \frac{\text{Equity}_{it}}{\text{Liabilities}_{it}}$$

The input data to calculate the value of EDR is taken by the author from the published financial reports of securities companies provided by Fiinpro.

Return-risk balancing capability (RMC):

As presented in the above sections, there is a trade-off relationship between increased profits and correspondingly increased bankruptcy risk, according to

Berger and Di Patti (2005), Akhtar et al. (2021), Nguyen Thuy Anh and Tran Thi Phuong Thao (2019). The fact that securities companies continue to expand their business activities also leads to increased bankruptcy risks, so these companies have to balance the trade-off between bankruptcy risk and profitability. In addition, Hung T. Nguyen et al. (2022), Bertrand and Schoar (2003) and Salvi et al. (2024) also showed that it is possible to both control bankruptcy risk and improve profitability, so the author decided to refer to previous studies by Tien Cao Minh et al. (2024) and Tran Van Hai (2022), Altman (1968) and the DuPont conceptual framework, the author proposes to use a new variable:

$$RMC = \left(\frac{\text{Equity}_{it}}{\text{Total Assets}_{it}} \right) * \left(\frac{\text{Working Capital}_{it}}{\text{Total Assets}_{it}} \right) * \left(\frac{\text{Profit after tax}_{it}}{\text{Revenue}_{it}} \right)$$

The input data to calculate the value of RMC is taken by the author from the published financial reports of securities companies provided by Fiinpro.

The RMC variable is a combination of 3 factors, including: the capital safety level of the enterprise, the ability to pay and mature debts, the ability to generate revenue and the efficiency of the enterprise. Securities companies must ensure the balance between the revenue generated (from increasing debt or expanding business areas to riskier areas) and the bankruptcy risk of the enterprise.

Asset utilization efficiency (AT):

Asset utilization efficiency plays a vital role in the operation and performance of a business. This ratio represents the ability to optimize the business and operational activities of the business. A higher AT means that the company is using assets effectively, conversely, a low AT shows that the company has not optimized the value of existing assets.

As cited in the conclusions of Altman (1968), Delen et al. (2013), Pouraghajan and Malekian (2012), Onaolapo and Kajola (2010), asset utilization efficiency is one of key factors contributing to a company's profitability and

bankruptcy risk is the efficiency of asset utilization. The formula of the AT variable is as follows:

$$AT = \frac{\text{Revenue}_{it}}{\text{Average Total Assets}_{it}}$$

The input data to calculate the value of AT is taken by the author from the published financial reports of securities companies provided by Fiinpro.

GDP Growth (GDP):

As cited from the conclusion of Ngan Bich Nguyen & Nhi Yen Vu (2025), Mkadmi et al. (2022), Yüksel et al. (2018), Egbunike and Okerekeoti (2018), besides the size of the enterprise, another key factor contributing to the profitability and bankruptcy risk of the enterprise is GDP growth. The formula of the GDP variable is as follows:

$$GDP = \frac{GDP_t - GDP_{t-1}}{GDP_{t-1}}$$

The input data to calculate the value of GDP is taken by the author from the published financial reports of securities companies provided by GSO.

Table 3.1. Summary of calculation formula, previous research results and expected signs of variables

Variable	Symbol	Expected sign	Formula
Dependent variable: ZSCORE			
Profitability	ROA	-	$ROA = \frac{\text{Net income}_{it}}{\text{Total Assets}_{it}}$
Company size	SIZE	-	$SIZE = \ln(\text{Total Assets}_{it})$
Equity-to-Debt ratio	EDR	+	$EDR = \frac{\text{Equity}_{it}}{\text{Total Liabilities}_{it}}$
Asset utilization efficiency	AT	+	$AT = \frac{\text{Revenue}_{it}}{\text{Average Total Assets}_{it}}$
Return-risk balancing capability	RMC	+	$RMC = \left(\frac{\text{Equity}_{it}}{\text{Total Assets}_{it}} \right) * \left(\frac{\text{Working Capital}_{it}}{\text{Total Assets}_{it}} \right) *$

			$\left(\frac{\text{Profit after tax}_{it}}{\text{Revenue}_{it}}\right)$
GDP growth	GDP	+	$\text{GDP} = \frac{\text{GDP}_t - \text{GDP}_{t-1}}{\text{GDP}_{t-1}}$
Dependent variable: ROA			
Bankruptcy risk	ZSCORE	-	$\text{ZSCORE}_{it} = 1.2 \text{ X1} + 1.4 \text{ X2} + 3.3 \text{ X3} + 0.6 \text{ X4} + 1.0 \text{ X5}$
Company size	SIZE	+	$\text{SIZE} = \ln(\text{Total Assets}_{it})$
Equity-to-Debt ratio	EDR	+	$\text{EDR} = \frac{\text{Equity}_{it}}{\text{Total Liabilities}_{it}}$
Asset utilization efficiency	AT	+	$\text{AT} = \frac{\text{Revenue}_{it}}{\text{Average Total Assets}_{it}}$
Return-risk balancing capability	RMC	+	$\text{RMC} = \left(\frac{\text{Equity}_{it}}{\text{Total Assets}_{it}}\right) * \left(\frac{\text{Working Capital}_{it}}{\text{Total Assets}_{it}}\right) * \left(\frac{\text{Profit after tax}_{it}}{\text{Revenue}_{it}}\right)$
GDP growth	GDP	+	$\text{GDP} = \frac{\text{GDP}_t - \text{GDP}_{t-1}}{\text{GDP}_{t-1}}$

3.4. Research data

Research data of 22 prominent securities companies operating in Vietnam was collected by the author within 14 years, from 2011 to 2024. Data of the above variables are collected from financial reports published on the websites of securities companies, or sources that the author obtained from Fiinpro, GSO, and financial reports.

3.5. Implementation process of the research model

Step 1: Calculating variables in the model

The author calculates the values of the variables using tools from Excel and puts them into Stata to prepare for step 2.

Step 2: Descriptive statistics of variables

Stata 17 software will be used to perform descriptive statistics for the dataset of 22 securities companies during the period from 2011 to 2024. In this section, the author will comment and evaluate input data such as: number of observations, average value, sample size, maximum value, minimum value...

Step 3: Correlation matrix analysis

In the next step, the author analyzes the correlation of variables in the model, examines the relationship between independent variables and dependent variables, and also identifies factors such as multicollinearity. Variables with a correlation higher than 0.85 indicate that the variables are showing signs of high multicollinearity and need to be carefully considered to help the research results of the model achieve high practical value.

Step 4: Panel Data Regression

In the research model, the author implements three main panel data regression methods, including Fix Effects Model (FEM) and Random Effects Model (REM) and System Generalized Method Of Moments (SGMM).

Step 5: Testing to choose the appropriate model

To determine the optimal regression model, the author performs the Hausman test. In which, H_0 : There is no correlation between the error term and the independent variable (choose REM) and H_1 : There is correlation (choose FEM). If $p\text{-value} < 0.05$, choose FEM, vice versa choose REM.

Step 6: Test for model defects

After selecting the model, the author performs the following tests:

Multicollinearity: Use the VIF index. If $VIF > 10$, the independent variable may cause multicollinearity.

Autocorrelation: The Wooldridge test is used. If $p\text{-value} < 5\%$, autocorrelation exists.

Heteroskedasticity:

REM: Breusch–Pagan test

FEM: Wald test

If $p\text{-value} < 5\%$, the model has a heteroskedasticity problem.

Step 7: Fix the model's defects by using SGMM:

Tests to assess the suitability of S-GMM include:

AR(1) and AR(2) tests:

AR(1): $p\text{-value} < 5\%$ (first-order autocorrelation expected).

AR(2): $p\text{-value} > 5\%$ (no second-order autocorrelation).

The Hansen–Sargan test: If the $p\text{-value} > 10\%$ shows that the instrument is suitable and the model effectively handles the endogeneity problem.

CHAPTER 3 SUMMARY

Chapter 3 presents in detail the research methods used in the thesis to achieve the research objectives. First, this chapter describes the data collection method, including primary and secondary data sources, scope and time of data collection. Next, the chapter presents the research variables, how to measure the variables and build a research model suitable for the analysis objectives.

The next part focuses on the data processing and analysis method, using statistical tools and techniques such as hypothesis testing, multivariate regression analysis, and econometric models suitable for the characteristics of panel data. In addition, the chapter also mentions the steps to check the reliability and validity of the research model, ensuring the analysis results are highly accurate.

In addition, the chapter clearly states the limitations of the research method and remedies to increase the objectivity and reliability of the research results.

CHAPTER 4: FINDINGS AND DISCUSSION

4.1. Research implementation

4.1.1. Descriptive statistics

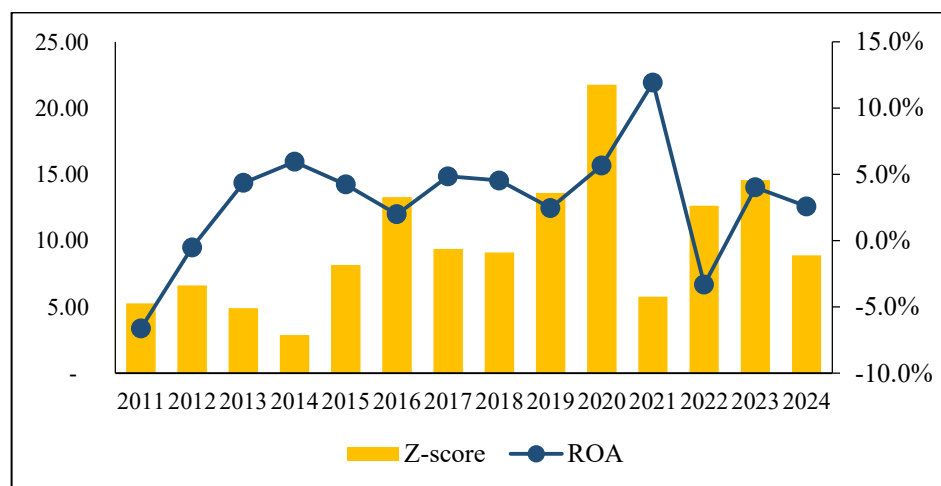
Table 4.1. Statistical results from variables in the model

Variable	Obs	Mean	Std. dev.	Min	Max
ZSCORE	308	9.768	17.085	-3.245	112.564
ROA	308	0.030	0.088	-0.347	0.556
SIZE	308	12.158	0.640	10.855	13.866
EDR	308	18.232	40.979	-0.088	342.328
AT	308	0.170	0.108	0.015	0.868
RMC	308	0.021	0.444	-3.212	0.794
GDP	308	0.060	0.015	0.026	0.080

Source: Stata 17

Bankruptcy risk (ZSCORE): The average bankruptcy risk of the securities companies in the research sample is 9.77. The lowest Z-score ever recorded was -3.245 at SBS Securities Company in 2015, while the highest index ever recorded was 112.564 at Wall Street Securities Company in 2020. During the 14 years from 2011 to 2024, the relationship between the bankruptcy risk (ZSCORE) and the profitability (ROA) of the companies reflected a clear positive relationship, except for 2012, 2020, and the period 2023 - 2024.

Image 4.1. Z-score and ROA of securities companies over the years



Source: compiled by author

Profitability (ROA): On average, securities companies recorded a ROA of about 3% over the 14-year period from 2011 to 2024. Although some companies recorded a RoA that deviated quite a bit from the average, such as Asia-Pacific Securities Joint Stock Company, which reached 55.61% in 2021 - the highest among the observations, and at the same time, this company also recorded a record low RoA of about -34.71% in 2022, in general, the value of the ROA variable still followed the normal distribution law, showing the reliability of the research data.

Company size (SIZE): The company size variable is measured by taking the natural logarithm of the company's total assets in the year under consideration. The average value of SIZE is about 12.158, with the largest value being 13.866, belonging to SSI Securities Corporation in 2024. The smallest value is 10.855, belonging to Tien Phong Securities Corporation in 2018.

Equity-to-Debt ratio (EDR): The EDR variable has an average value of about 18.231, on the contrary, considering the median value of EDR is only 2.512. This is because there are many securities companies in the sample set with EDR much higher than the remaining samples, leading to a change in the average mark, especially for small-scale securities companies. However, according to the distribution chart, most of the values of the EDR variable still revolve around the normal distribution. Among them, the company that recorded the highest EDR value was Hoa Binh Securities Joint Stock Company, with an EDR variable value of 343.328 in 2018. On the contrary, SBS Securities Joint Stock Company recorded a record low EDR variable value in 2012 with only -0.088.

Asset utilization efficiency (AT): The asset utilization efficiency of the companies in the sample set is close to 0.17. In particular, the majority of companies have an asset utilization efficiency level of 0.015 to 0.868. Tien Phong Securities Joint Stock Company recorded the highest value of AT, about 0.868 in 2019. On the contrary, this company also recorded the lowest value of AT, reaching a value of about 0.0148 in 2013.

Return-risk balancing capability (RMC): The RMC variable represents the company's ability to balance risk and return in the trade-off relationship between the company's bankruptcy risk and profitability. The data shows that the average RMC value is 0.021. In which, the company with the highest RMC value of 0.794 is APG Securities Joint Stock Company in 2014. The company with the lowest RMC value is Vietnam Financial Investment Securities Joint Stock Company in 2019 with a value of -3.213.

GDP Growth (GDP): The GDP variable recorded an average value of 6.01%, showing that the economic growth rate has always been high in the period from 2011 to 2024. The GDP variable recorded the highest value in 2022 with a gross domestic product growth rate of 8.02% due to the strong economic recovery after the pandemic due to the open-door policy and recovering demand. In contrast, the GDP variable recorded the lowest value in 2.58% in 2021 due to the COVID-19 pandemic outbreak causing supply chain disruption, production stagnation, and limited trading activities.

4.1.2. Correlation coefficient analysis

Evaluation of the correlation matrix of the research data shows that the variables in the model do not have a very high correlation with each other. Except for the EDR variable for ZSCORE, which recorded a value of 0.813, the highest among the remaining pairs of variables, followed by the pairs of RMC variables with ROA, ROA with AT and ROA with SIZE. On the contrary, the pair of variables that recorded the smallest value was between SIZE and ZSCORE, reaching -0.317.

Table 4.2. Correlation coefficient analysis

Variable	ZSCORE	ROA	SIZE	EDR	AT	RMC	GDP
ZSCORE	1.000						
ROA	-0.132	1.000					
SIZE	-0.317	0.204	1.000				
EDR	0.813	-0.152	-0.306	1.000			
AT	-0.144	0.440	0.190	-0.275	1.000		

RMC	-0.087	0.638	0.111	-0.077	0.184	1.000	
GDP	-0.056	-0.274	-0.092	0.028	-0.147	-0.168	1.000

Source: Stata 17

4.1.3. Estimating regression models using FEM, REM, SGMM models

4.1.3.1. Regression results using FEM, REM, SGMM for ZSCORE model

Table 4.3. Panel data regression analysis using FEM, REM, SGMM for dependent variable ZSCORE

	FEM	REM	SGMM
ROA	-3.166 (-0.33)	-5.911 (-0.63)	-65.155*** (-2.69)
SIZE	-3.981*** (-4.09)	-3.335*** (-3.53)	-3.189** (-2.11)
EDR	0.325*** -22.27	0.330*** -22.89	0.382*** -3.86
AT	14.445*** -2.4	15.649*** -2.65	25.559* -1.92
RMC	-0.541 (-0.33)	-0.573 (-0.36)	14.078** -2.37
GDP	0 (.)	-96.421 (-1.56)	-157.311** (-2.22)
_cons	49.907*** -4.17	47.633*** -3.87	51.205** -2.2
R ²	0.672	0.682	
Observations	308	308	286

values are shown in parentheses; ***, **, * represent statistical significance levels of 1%, 5% and 10%, respectively

Source: Stata 17

The regression results according to the three methods FEM, REM, SGMM are presented in Table 4.3 showing that the R² coefficients of the FEM and REM models are 0.672 and 0.682 respectively, indicating that the independent variables

have a fairly high explanatory power, reaching 67.2% and 68.2% for the dependent variable of bankruptcy risk (ZSCORE). In all three models, independent variables such as company size (SIZE), equity-to-Debt ratio (EDR) and asset utilization efficiency (AT) all show statistically significant results for the independent variable ZSCORE. While the remaining variables such as profitability (ROA), return-risk balancing capability (RMC) and GDP growth (GDP) are only statistically significant in the SGMM model. Overall, the regression results from the three models show that firm size (SIZE) and equity-to-debt ratio (EDR) have a significant impact on the bankruptcy risk of securities companies at the significance level from 1% to 5%.

4.1.3.2. Regression results using FEM, REM for ROA model

Table 4.4. Panel data regression analysis using FEM, REM, SGMM for dependent variable ROA

	FEM	REM	SGMM
ZSCORE	0.000 (-0.33)	0.000 (-0.50)	-0.002* (-1.83)
SIZE	0.012* -1.930	0.011* -1.880	0.012* -1.930
EDR	0.000 -0.060	0.000 -0.230	0.001 -1.420
AT	0.239*** -7.000	0.242*** -7.180	0.294*** -6.690
RMC	0.102*** -13.000	0.103*** -13.320	0.099*** -3.350
GDP	0.000 (.)	-0.795 (-1.57)	-0.671** (-2.26)
_cons	-0.153** (-2.06)	-0.099 (-1.23)	-0.109 (-1.36)

R^2	0.521	0.538	
Observations	308	308	286

*values are shown in parentheses; ***, **, * represent statistical significance levels of 1%, 5% and 10%, respectively*

Source: Stata 17

The regression results according to the three methods FEM, REM, SGMM are presented in Table 4.4, showing that the R^2 coefficients of the FEM and REM models are 0.521 and 0.538, respectively, equivalent to the explanation level of about 52.1% and 53.8% of the independent variables for the profitability of securities companies. In all three models, independent variables such as company size (SIZE), asset utilization efficiency (AT) and return-risk balancing capability (RMC) all show statistically significant results. For the remaining variables such as GDP growth (GDP) and bankruptcy risk (ZSCORE), they are only statistically significant for the SGMM model. In all three models, AT and RMC have high statistical significance at the 1% level, while the independent variable EDR does not reach statistical significance in all three models.

4.1.3.3. Testing for model defects

Hausman testing:

The level of fit is performed through Hausman test, the test is performed and the conclusion is made through 2 hypotheses:

H_0 : REM model is suitable

H_1 : FEM model is suitable

The author conducts Hausman test with bankruptcy risk model and gets the following results:

Table 4.5. Hausman test for bankruptcy risk model

Chi2(5) =	8.48
Prob > chi2 =	0.132

Source: Stata 17

After the Hausman test result with Prob > chi2 result of 0.132 higher than 5%, the author accepts the hypothesis H_0 : there is no correlation between

independent variables and random errors. Therefore, the appropriate model to describe the regression coefficients is the REM model (Random Effects Model).

The REM model, with an R-squared value of 0.682, shows that the variables in the model have significantly explained the fluctuations of ZSCORE. Most of the variables in the REM model have a significance level below significant (from 10% to 1%), except for the variables ROA, RMC, GDP, which have not shown the ability to significantly impact the dependent variable ZSCORE. In the list of statistically significant variables, EDR and AT are showing signs of fluctuating in the same direction with the dependent variable ZSCORE. When EDR and AT increase, the ZSCORE variable also increases at a corresponding rate. On the contrary, the variable SIZE shows signs of opposite fluctuations to ZSCORE.

The author conducts Hausman test with profitability risk model and gets the following results:

Table 4.6. Hausman test for profitability model

Chi2(5) =	2.57
Prob > chi2 =	0.766

Source: Stata 17

After the Hausman test result with Prob > chi2 result of 0.766 higher than 5%, the author accepts the hypothesis H_0 : there is no correlation between independent variables and random errors. Therefore, the appropriate model to describe the regression coefficients is the REM model (Random Effects Model).

The REM model, with an R-squared value of 0.538, shows that the variables in the model have significantly explained the fluctuations of ROA. Most of the variables in the REM model have a significance level below significant (from 10% to 1%), except for the variables ZSCORE, EDR, and GDP, which have not shown a significant impact on the dependent variable ROA. In the list of statistically significant variables, AT, SIZE, and RMC are showing signs of fluctuating in the same direction with the ROA.

Multicollinearity testing:

The author performed a VIF test to check for multicollinearity in the regression model. In the case of the dependent variable ZSCORE, the test results showed that the model did not have multicollinearity, so no variables needed to be removed. Specifically, the average VIF value of the regression model is 1.44, the VIF values of the variables in the model ranged from 1.08 to 2.16, all of which were less than 10.

Table 4.7. Multicollinearity test for models with dependent variable ZSCORE

Dependent variable: ZSCORE		
Variable	VIF	1/VIF
ROA	2.16	0.463
RMC	1.72	0.581
AT	1.34	0.745
EDR	1.17	0.856
SIZE	1.14	0.877
GDP	1.08	0.922
Mean VIF	1.44	

Source: Stata 17

The author conducted a VIF test to check for multicollinearity in the regression model. In the case of the dependent variable ROA, the test results showed that the model did not have multicollinearity, so there was no need to remove any variables. Specifically, the average VIF value of the regression model is 1.81, the VIF values of the variables in the model varied from 1.07 to 3.23, all of which were less than 10.

Table 4.8. Multicollinearity test for models with dependent variable ROA

Dependent variable: ROA		
Variable	VIF	1/VIF
EDR	3.23	0.309
ZSCORE	3.15	0.318
AT	1.17	0.857
SIZE	1.15	0.867
GDP	1.07	0.935
RMC	1.07	0.936

Mean VIF	1.81
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Source: Stata 17

Autocorrelation testing:

For the case of the dependent variable ZSCORE, the Wooldridge test shows that $\text{Prob} > F = 0.188$, so H_0 cannot be rejected, meaning there is no evidence of first-order autocorrelation. In addition, in the case of the dependent variable RoA, $\text{Prob} > F = 0.001$ is less than the 5% threshold, so H_0 is rejected, indicating that the model has first-order autocorrelation, which needs to be adjusted to ensure the reliability of the estimate.

Table 4.9. Autocorrelation test table for 2 models

Dependent variable:	ZSCORE	ROA
Wooldridge Test	$F(1,21) = 1.850$	$F(1,21) = 14.764$
	$\text{Prob} > F = 0.188$	$\text{Prob} > F = 0.001$
Conclusion	Fail to reject $H_0 \rightarrow$ No first-order autocorrelation	Reject $H_0 \rightarrow$ Evidence of first-order autocorrelation

Source: Stata 17

Heteroscedasticity testing:

For the Breusch and Pagan Lagrangian test for heteroscedasticity, both models showed similar results. $\text{Prob} > \chi^2 = 0.000$, less than the 5% significance level shows that both models have heteroscedasticity.

Table 4.10. Heteroscedasticity test table for 2 models

Dependent variable:	ZSCORE	ROA
Breusch - Pagan Test	$\text{Chibar}(01) = 19.98$	$\text{Chibar}(01) = 50.77$
	$\text{Prob} > \chi^2 = 0.000$	$\text{Prob} > \chi^2 = 0.000$
Conclusion	Reject $H_0 \rightarrow$ heteroskedasticity detected	Reject $H_0 \rightarrow$ heteroskedasticity detected

Source: Stata 17

4.1.3.4. Addressing defects using the SGMM model

Addressing defects in the model with ZSCORE as the dependent variable:

Table 4.11. SGMM model results table with dependent variable ZSCORE

Dependent variable: ZSCORE		
Variable	Hệ số β	P > z
L1.ZSCORE	-0.261**	0.014
ROA	-65.155***	0.007
SIZE	-3.189**	0.034
EDR	0.382***	0.000
AT	25.559*	0.055
RMC	14.078**	0.018
GDP	-157.311**	0.026
Constant	51.205**	0.028
Number of instruments = 14		
Number of groups = 22		
Prob > chi2 = 0.000		
Arellano-Bond test for AR (1) in first differences: z = -1.53		Pr > z = 0.126
Arellano-Bond test for AR (2) in first differences: z = -1.92		Pr > z = 0.055
Hansen test of overid. restrictions: chi2(6) = 2.22		Prob > chi2 = 0.898

*values are shown in parentheses; ***, **, * represent statistical significance levels of 1%, 5% and 10%, respectively*

Source: Stata 17

As presented before, in order to overcome problems such as autocorrelation, heteroscedasticity, the author has run the SGMM model to overcome the shortcomings of the regression model with the dependent variable ZSCORE. From the results, it shows that the number of instruments (which is 14) is lower than the number of groups (which is 22). Therefore, the model does not have the Overlifting phenomenon. Besides, the Arellano-Bond test for AR (2) in first differences $Pr > z = 0.055$, higher than 5%, so the model does not have the second-order

autocorrelation phenomenon. In addition, the Hansen test of overid. restrictions $\text{Prob} > \chi^2 = 0.898$ higher than 10%. Therefore, H_0 is accepted: the instrumental variable is appropriate. Thus, the model has overcome the endogeneity problem. With the above test results, the SGMM model is qualified to be accepted as a suitable and reliable model. In summary, the results from the SGMM model show that the dependent variable ZSCORE is affected by 6 independent variables, including ROA, SIZE, EDR, AT, RMC, and GDP.

Bankruptcy risk regression model:

$$\text{ZSCORE} = -65.155 \text{ ROA} - 3.189 \text{ SIZE} + 0.382 \text{ EDR} + 25.559 \text{ AT} + 14.078 \text{ RMC} - 157.311 \text{ GDP} + 51.205$$

Addressing defects in the model with ROA as the dependent variable:

Table 4.12. SGMM model results table with the dependent variable as ROA

Dependent variable: ROA		
Variable	Hệ số β	P > z
L1.ROA	-0.141	0.133
ZSCORE	-0.002*	0.068
SIZE	0.012*	0.054
EDR	0.001	0.157
AT	0.294***	0.000
RMC	0.099***	0.001
GDP	-0.671**	0.024
Constant	-0.109	0.175
Number of instruments = 13		
Number of groups = 22		
Prob > $\chi^2 = 0.000$		
Arellano-Bond test for AR (1) in first differences: z = -2.03		Pr > z = 0.043
Arellano-Bond test for AR (2) in first differences: z = -1.48		Pr > z = 0.138
Hansen test of overid. restrictions: $\chi^2(5) = 3.81$		Prob > $\chi^2 = 0.577$

*P-values are shown in parentheses; ***, **, * represent statistical significance levels of 1%, 5% and 10%, respectively*

Source: Stata 17

As presented before, in order to overcome problems such as autocorrelation, heteroscedasticity, the author has run the SGMM model to overcome the shortcomings of the regression model with the dependent variable ROA. From the

results, it shows that the number of instruments (which is 13) is lower than the number of groups (which is 22). Therefore, the model does not have the Overlifting phenomenon. Besides, the Arellano-Bond test for AR (2) in first differences $\text{Pr} > z = 0.138$ is higher than 5%. Therefore, the model does not have the second-order autocorrelation phenomenon. In addition, the Hansen test of overid. restrictions $\text{Prob} > \chi^2 = 0.577$, higher than 10%. Therefore, H_0 is accepted: the instrumental variable is appropriate. Thus, the model has overcome the endogeneity problem. With the above test results, the SGMM model is qualified to be accepted as a suitable and reliable model. In summary, the results from the SGMM model show that the dependent variable ROA is affected by 5 independent variables, including ZSCORE, SIZE, AT, RMC, GDP, except EDR.

Profitability regression model:

$$\text{ROA} = -0.002 \text{ ZSCORE} + 0.012 \text{ SIZE} + 0.001 \text{ EDR} + 0.294 \text{ AT} + 0.099 \text{ RMC} - 0.671 \text{ GDP} - 0.109$$

4.1.3.5. Hypothesis testing

For bankruptcy risk model:

Among the factors affecting ZSCORE, the variables ROA, SIZE, EDR, AT, RMC, GDP are all statistically significant. The regression results of the variables are consistent with the initial hypothesis, except GDP.

Table 4.13. Testing research hypotheses on factors affecting bankruptcy risk

Dependent variable: ZSCORE				
Variable	Expected Sign	Research Result	Significance Level	Hypothesis Test Result
ROA	(-)	(-)	1%	Accepted
SIZE	(-)	(-)	5%	Accepted
EDR	(+)	(+)	1%	Accepted
AT	(+)	(+)	10%	Accepted

RMC	(+)	(+)	5%	Accepted
GDP	(+)	(-)	5%	Rejected

Source: compiled by author

For profitability model:

Among the factors affecting ROA, the variables ZSCORE, SIZE, AT, RMC, GDP are all statistically significant, only one variable, EDR, has not achieved statistical significance. The regression results of the variables are consistent with the initial hypothesis, except GDP.

Table 4.14. Testing research hypotheses on factors affecting profitability

Dependent variable: ROA				
Variable	Expected Sign	Research Result	Significance Level	Hypothesis Test Result
ZSCORE	(-)	(-)	10%	Accepted
SIZE	(+)	(+)	10%	Accepted
EDR	(-)	Not statistically significant		
AT	(+)	(+)	1%	Accepted
RMC	(+)	(+)	1%	Accepted
GDP	(+)	(-)	5%	Rejected

Source: compiled by author

4.2. Discussion

4.2.1. For the relationship between bankruptcy risk and profitability

The results from the SGMM model show that the bilateral relationship between bankruptcy risk and the profitability of securities companies is a positive correlation. In which profitability (ROA) has a positive impact on bankruptcy risk

(ZSCORE) with a regression coefficient of -65.155 and bankruptcy risk (ZSCORE) also showed a positive impact on profitability (ROA) with a regression coefficient of -0.002. The positive relationship between bankruptcy risk and profitability of securities companies shows that if the bankruptcy risk of securities companies is higher (the value of Z-score is low), then the profitability of securities companies will increase accordingly. This can be explained according to Agency cost theory that the managers of securities companies are under more financial pressure and motivation, these managers are forced to act carefully and rationally to improve the profitability of the company to avoid bankruptcy. Conversely, if bankruptcy risk is controlled and maintained at a low level (the Z-score value remains high) managers are not under pressure or motivated to improve the company's profitability.

Specifically, the bidirectional relationship between bankruptcy risk and profitability is also shown through the SIZE variable. By increasing the size of the company, securities companies will have an advantage in attracting customers, investing in human resources, improving brand recognition and corporate reputation, etc. However, a larger size increases the risk of bankruptcy. Due to the heavy dependence on short-term capital to finance business operations, increased size puts more pressure on the company to meet the growing financial demand. In addition, a larger size makes the company's proprietary trading activities more difficult due to the need to manage a larger and more complex volume of assets and investment portfolios. The regression results show that GDP increases bankruptcy risk and reduces corporate profitability. This places pressure on managers to be more cautious and to continuously build and improve their risk management systems.

However, the regression results from the SGMM model also show that the ability to optimize the company's resources, the ability to optimize the company's profits while ensuring a stable capital structure, and controlled bankruptcy risks will play an important role in the sustainable development of the company, helping the company improve its profitability and simultaneously control and limit the

company's bankruptcy risk. This result is similar to the Agency cost theory, in the situation of increased bankruptcy risk, financial obligations will create a strong motivation for managers to act cautiously and appropriately to improve and enhance the company's profitability. On the contrary, when financial pressures and bankruptcy risks are controlled and maintained at a low level, the motivation to promote managers to act cautiously will no longer be significant. Therefore, the profitability of companies will be reduced.

4.2.2. For bankruptcy risk model

4.2.2.1. Profitability (ROA)

In line with the author's expectation, the SGMM model results show that profitability has a significant impact on the variation of bankruptcy risk - there is a trade-off relationship between profit and risk in the operations of securities companies with a significance level of 1%, the coefficient β is -65.155 .

In the sample collected by the author, the securities company with the clearest relationship between profitability and bankruptcy risk is Agribank Securities Joint Stock Company (Agriseco). In the period from 2015 to 2016, the company's ZSCORE variable reached a record high of 21.655 and 73.082, but the company's ROA in this period was also at a record low, reaching -7.801% and -21.695% . In the period from 2017 to 2024, the company's ZSCORE gradually decreased from 49.510 to 3.436, and the company's ROA gradually increased from 3.798% to 4.136% . Similar to AGR, many other companies also clearly show the positive relationship between ZSCORE and ROA, such as VND, IVS, VIX, etc.

For companies with a high debt structure, profitability tends to be higher than other companies. This can be explained by the fact that corporate managers are subject to additional supervision and pressure from financial leverage, which forces them to improve and enhance the company's operational efficiency in order to meet debt repayment obligations, maintain liquidity, and protect their reputation.

4.2.2.2. Company size (SIZE)

The research results for the bankruptcy risk model show that the variable company size (SIZE) reaches statistical significance at a significance level of 5% for the bankruptcy risk of securities companies with a regression coefficient β of -3.189, consistent with the initial research hypothesis.

The scale of securities companies is generally not too large; the average total asset value of companies in the sample is about 12 trillion, and the median is 6 trillion VND. Among them, there are a few companies with a total asset value that is much higher than other companies, affecting the average value, such as SSI (73 trillion VND), VND (44 trillion VND), and HCM (31 trillion VND). Therefore, a typical example to clearly illustrate the inverse relationship between company size and bankruptcy risk can be mentioned in the case of the two most prominent securities companies in the market today, including SSI Securities Corporation, VNDIRECT Securities Corporation, and Ho Chi Minh City Securities Corporation. During 14 years of continuous growth in total assets, with CAGRs of 18.91%, 27.43%, and 19.37%, respectively, the ZSCORE ratio of the above two securities companies decreased, with CAGRs of -7.07%, -3.48%, and -8.45%, respectively.

Although the large size of a company can create a significant competitive advantage for all businesses due to access to cheap capital, high-quality human resources, reputation, and the ability to attract customers. However, for securities companies, being too large in scale brings significant risks. Firstly, in terms of capital structure, most securities companies often depend on short-term capital while still holding long-term investments. This puts pressure on companies to closely manage solvency risk and bankruptcy risk due to the maturity mismatch between short-term funding and long-term investments. In the event of market fluctuations, companies with large positions will find it difficult to act promptly, which can impact business operations and increase expenses, leading to an increase in bankruptcy risk.

4.2.2.3. Equity-to-Debt ratio (EDR)

The research result for the bankruptcy risk model shows that the variable Equity-to-Debt ratio (EDR) is statistically significant at a 1% significance level for the stability of securities companies with a regression coefficient β of 0.382, consistent with the research hypothesis.

It is undeniable that debt capital helps companies improve their competitiveness and increase profits when expanding their business operations. However, due to the nature of depending on short-term capital and the riskiness of the stock market, borrowing will reduce the company's ability to absorb inherent risks when the market fluctuates, especially for companies under pressure on liquidity. Therefore, many companies, especially small-scale securities companies, often choose to use equity capital instead of borrowed capital - due to lack of access to cheap capital and management capacity not being competitive enough with larger securities companies, as the main source of capital to finance business operations, leading to the fact that equity capital at these companies is often many times higher than the debt portion.

4.2.2.4. Asset utilization efficiency (AT)

For the bankruptcy risk model, AT reaches a regression coefficient value of 25.559 with a significance level of 10%, also consistent with the research hypothesis.

The AT variable shows the ability to optimize the capital of the enterprise during the business operation process. The higher the AT ratio, the more the company is optimizing its business model, limiting the waste of resources. This is especially important in the financial sector, when most of the company's assets are recorded in investments or loans. Improving the AT ratio helps the company reduce bankruptcy risk, increase stability and limit liquidity risks or bankruptcy because the company has become too large and it is difficult to control and interact with the company's investments or loans... The improved and controlled AT ratio shows that securities companies are utilizing assets effectively. This helps improve the

company's cash flow, helping to meet liquidity needs or pay debts, thus the bankruptcy risk can be controlled.

4.2.2.5. Return-risk balancing capability (RMC)

For the model with ZSCORE as the dependent variable, the RMC reaches a regression coefficient value of 10.623 with a significance level of 10%. The result is within the author's expectations.

Some companies often face two choices, one is lower profitability risk for lower bankruptcy risk or higher risk of bankruptcy with higher profit. This problem is common to many companies, including companies operating in the stock market in Vietnam. Being an industry with fierce competition and high risks, the issue of choosing higher profits or lower bankruptcy risk is often weighed, planned, and measured in detail by many companies. If choosing profits, there will be an increase in the risk of bankruptcy, if choosing stability, it may lose the ability to grow. Therefore, companies in the industry often have to spend a lot of resources to build and standardize risk management standards, train high-quality employees, to achieve lower bankruptcy risk and higher profitability.

4.2.2.6. GDP growth (GDP)

The results from the SGMM model show that the GDP variable harms the ZSCORE ratio of securities companies with a coefficient value of -157.312 and a significance level of 5%. This is contrary to the initial hypothesis that GDP can help companies control and limit bankruptcy risks.

In general, GDP growth can make the bankruptcy risk higher, mostly from the increase of interest rates on short-term loans. GDP growth shows that production and business activities are starting to recover, and can also lead to an increase in interest rates. Due to the high dependence on short-term loans, the increase in interest rates will directly affect the financial structure of securities companies, causing increased financial pressure and increasing bankruptcy risks. In addition, when the economy recovers, capital flows away from the financial market

and into production and business activities, which will affect the income of securities companies. Declining earnings put additional pressure to offset rising interest rates, causing bankruptcy risk to increase.

4.2.3. For profitability model

4.2.3.1. Bankruptcy risk (ZSCORE)

In line with the author's expectations, the SGMM model results show that bankruptcy risk (ZSCORE) has a significant impact on the profitability (ROA) and there is a trade-off relationship between profit and bankruptcy risk in the operations of securities companies with a significance level of 10%, the coefficient β is -0.002.

Z-score is a measure of a company's bankruptcy risk. A high Z-score indicates a low bankruptcy risk and a low Z-score indicates a high bankruptcy risk. Securities companies that plan to reduce bankruptcy risk should limit their use of leverage, reduce borrowing, expand capital, and expand investment activities into safer and more stable financial products. However, a company with a very high Z-score or low bankruptcy risk must accept that its profitability is limited and at the same time face a significant opportunity cost in profits compared to riskier competitors.

4.2.3.2. Company size (SIZE)

The research results for the model with ROA as the dependent variable show that the variable company size (SIZE) reaches statistical significance at a significance level of 10% for the bankruptcy risk of securities companies with a regression coefficient β of 0.012, consistent with the research hypothesis.

The size of a company reflects its position and reputation in the market and is the basis for measuring the company's competitiveness. Companies with large capital will be able to invest in expanding their business operations, building and training high-quality employees, improving their image and enhancing their position in the market. Larger scale also helps businesses easily attract cheap capital sources, attract customers and strategic partners in domestic and foreign markets.

4.2.3.3. Equity-to-Debt ratio (EDR)

The regression results from the SGMM model show that the Equity-to-Debt ratio variable does not show a statistically significant relationship with corporate profitability.

From the research data, the correlation between EDR and ROA of each company is somewhat different. Large and small securities companies such as SSI, VND, VIX, AGR, IVS recorded a negative correlation between EDR and ROA, while medium-sized securities companies such as ORS, BSI, CTS, TVS recorded a positive correlation.

For large securities companies such as SSI, VND, the diversified revenue structure of many business segments allows for increased borrowing to expand business operations. In addition, thanks to their reputation in the market, large securities companies can enjoy lower interest rates. Therefore, the ROA ratio can grow. For medium-sized securities companies such as ORS, BSI, CTS, TVS. These companies often have investment banking and proprietary trading activities accounting for a large proportion of their business activities, so increasing debt, especially short-term loans, will create financial pressure on the company. Therefore, reducing ROA. For small securities companies such as AGR and IVS, there is a negative correlation between EDR and ROA because these companies diversify their revenue segments but mainly have the majority of revenue from lending, so increasing debt can help the company make a profit from the interest rate difference, thereby increasing ROA.

4.2.3.4. Asset utilization efficiency (AT)

For the model with ROA as the dependent variable, AT variable reaches statistical significance at 1% with a β coefficient of 0.294, showing that asset utilization efficiency has a statistically significant impact on the profitability of securities companies, consistent with the research hypothesis.

Due to the fierce competition in the industry, many companies need to optimize their capital structure to reduce cost pressure. Some large companies often rely heavily on short-term debt, such as bank loans or short-term bonds, to finance their business operations. This short-term capital with lower costs will be used to finance their operations.

4.2.3.5. Return-risk balancing capability (RMC)

For the profitability model, the RMC reaches a regression coefficient value of 0.099 with a significance level of 1%. Both of these results are within the author's expectations.

For an industry that is constantly under high-risk competitive pressure like companies in the stock market, the ability to balance risk and profit will play an important role in the company's business operations. The ability to balance risk and profit can be considered through factors such as the capacity of the board of directors, risk management mechanisms and systems, the company's human resources, and modern technology. Investing in building a high-quality human resource team and new technologies such as AI allows the company to improve its profitability and limit the risk of bankruptcy.

4.2.3.6. GDP growth (GDP)

The results from the SGMM model show that the GDP variable has a negative impact on the ROA ratio of securities companies with a coefficient value of -0.671 and a significance level of 5%. This is contrary to the initial hypothesis that GDP has a positive impact on the profitability of companies.

This result is different from the author's expectations. However, in reality, the GDP growth index has not shown a high correlation with the profitability of securities companies but also depends on many other macro and micro factors. For example, in 2021, when the economy's supply chain was disrupted, production and trade were stagnant, the ROA of securities companies recorded a high growth rate. Four major securities companies such as SSI, VND, HCM, and MBS recorded

record high ROA of 6.23%, 9.15%, 6.22% and 6.50%, equivalent to growth of 55.71%, 76.10%, 17.21%, and 42.25% respectively, compared to the same period last year. This is due to the growth of trading activities in the stock market, when individual investors and institutional investors take advantage of idle money to invest in the context of the Covid outbreak, leading to social distancing.

In general, GDP growth has a negative impact on ROA because when the GDP growth rate is high, strong economic growth will attract investment cash flow to other investment channels instead of focusing only on the stock market, causing trading activities to decline. When the GDP growth rate reduce, the State Bank of Vietnam can lowers interest rates to stimulate the economy. Lower interest rates attract cash flow back to investment channels, causing the profitability (ROA) of securities companies to grow.

CHAPTER 4 SUMMARY

In chapter 4, the study conducted a comprehensive assessment of the bidirectional relationship between bankruptcy risk and profitability of 22 securities companies operating in Vietnam during the period from 2011 to 2024. In this chapter, the author used the FEM and REM regression models to evaluate and analyze this bidirectional relationship and performed tests such as multicollinearity, heteroscedasticity, endogeneity, etc., then ran the SGMM model to overcome the defects of the current models.

From the results obtained, it can be affirmed that the relationship between the bankruptcy risk and the profitability of securities companies is inverse. The research results are analyzed and discussed in detail, clearly, and closely related to reality.

Based on the results obtained in Chapter 4, the author will make recommendations on management implications in Chapter 5.

CHAPTER 5: CONCLUSION

5.1. Conclusion

The research results confirm that the bankruptcy risk of securities companies fluctuates positively with their profitability, with a statistical significance level of 10%. Similarly, the profitability also has a positive impact on the bankruptcy risk of these companies, with a significance level of 1%. This means that there is a real trade-off relationship between profit and bankruptcy risk for securities companies in Vietnam in the period 2011 – 2024.

In the relationship between bankruptcy risk and profitability of securities companies, an increase in bankruptcy risk will increase profitability, and conversely, a decrease in bankruptcy risk will reduce the profitability of securities companies. In addition, the company size factor also helps explain this relationship. Although a larger company size helps companies to have higher profitability, it also makes these companies face higher bankruptcy risk. Conversely, a smaller company size will help lower the company's bankruptcy risk but reduce the company's profitability.

In addition, the research results also show that companies should improve the management capacity of the company's board of directors. The regression results from the SGMM model show that if a company makes good use of resources, optimizes profits, and controls capital structure, it can help the company both control bankruptcy risk and improve profitability.

5.2. Managerial implications

Overall, based on the results obtained, the author proposes some implications for the executive boards of some securities companies to improve the performance and position of the enterprise:

Regarding capital policy, enterprises should prioritize increasing retained earnings and diversifying capital sources (issuing preferred shares, secured bonds,

strategic cooperation), while reasonably diversifying debt maturity to avoid refinancing risks.

Regarding incentive mechanisms and human resource management, the design of remuneration systems, equitization and performance bonuses should be closely linked to indicators such as ROA and Z-score. At the same time, a long-term bonus mechanism should be implemented.

Regarding operations and technology, applying data analysis technology, automation and optimizing asset utilization efficiency helps increase revenue per unit of asset without increasing leverage.

Regarding risk management, setting clear internal limits, conducting periodic stress tests, and developing response scenarios for systemic risks.

However, denying that securities companies pursue markedly different business models and growth strategies, some rely primarily on a single revenue stream such as proprietary trading, financial advisory, or investment banking. In contrast, others have built a diversified revenue mix encompassing brokerage, asset management, margin lending, and more. Managers must tailor and select recommendations to suit each company's unique characteristics.

Specifically, for companies reliant on a single income source, the focus should be on optimizing core operational processes and rigorously managing sector-specific risks, while gradually expanding complementary service lines to mitigate excessive dependence on a single channel. Conversely, companies with diversified revenue bases can implement measures to rebalance profit contributions across divisions, reinforce financial capacity in slower-growth segments, and leverage their strongest business lines to lower the overall cost of capital.

By adopting a flexible, selective approach to credit, liquidity, and market-risk controls that align with each organization's revenue structure, a securities company can not only enhance profitability but also strengthen its resilience against adverse market fluctuations.

5.2.1. Profitability (ROA)

The study finds that profitability is positively associated with bankruptcy risk. Accordingly, the author recommends the following measures to improve profitability while simultaneously containing bankruptcy risk:

Establish a remuneration, equity-ownership and performance-bonus framework tied to metrics such as ROA and Z-score to incentivize the behaviour of key personnel.

Strengthen internal oversight and the independent board by appointing additional independent directors, enhancing the quality of the audit committee and risk committee, and conducting regular internal audits.

5.2.2. Bankruptcy risk (ZSCORE)

The research results show that bankruptcy risk has an adverse impact on the company's profitability. Therefore, the author has some suggestions as follows to help the company reduce bankruptcy risk and improve profitability at the same time:

Increase retained earnings by adopting a cyclical dividend policy: retain more profits during favorable periods to finance high-return investments and reduce dependence on external borrowing.

When it is necessary to mobilize external funds, prioritize debt with low cost and suitable maturity; avoid using short-term debt to finance long-term assets; phase in debt maturity to reduce refinancing risk.

5.2.3. Company Size (SIZE)

In the context of the company's increasingly expanding scale, the board of directors needs to proactively develop a sustainable growth strategy, while closely controlling financial risks arising from increasing debt structure. Specifically, the company should:

Determine the optimal scale to maintain the financial leverage ratio within the safe threshold, and avoid overconfidence in expanding scale only through debt channels.

Diversify capital sources through the issuance of preferred shares, corporate bonds with collateral or capital mobilization from strategic partners, reduce short-term debt repayment pressure and avoid liquidity risks.

Establish monitoring mechanisms and contingency scenarios for growth slowdowns or interest-rate volatility, ensuring sufficient working-capital buffers and ready plans for debt restructuring if needed.

Develop a phased expansion plan aligned with expected profit targets and debt repayment capacity to avoid overextension when revenue growth requires resources exceeding the firm's funding capability.

5.2.4. Equity-to-Debt ratio (EDR)

Given that the Equity-to-Debt ratio has a negative impact on bankruptcy risk but has not yet had a statistically significant impact on profitability, management should develop the following strategies:

Avoid excessive debt growth leading to high interest expenses, putting pressure on profit margins even though bankruptcy risk is controlled.

Consider issuing preferred shares or convertible bonds, both to supplement equity and to keep borrowing costs at a controllable level.

5.2.5. Asset utilization efficiency (AT)

In the context of efficient use of assets both contributing to improving profitability and minimizing bankruptcy risks, company leaders need to consider this a strategic pillar of the company by:

Apply data analysis and automation technology to increase transaction processing speed and reduce capital turnover time.

Provide in-depth training for operational personnel to maximize the capacity of the existing trading system and facilities.

Develop additional value-added products and services (e.g. investment consulting, asset management services) using existing technology infrastructure to optimize costs on total assets.

Leveraging multiple channels from digital assets (online trading platforms, mobile applications) to diversify revenue streams without increasing fixed investments.

5.2.6. Return-risk balancing capability (RMC)

In the context of the ability to balance risk and profit showing a positive impact on both profitability and bankruptcy risk reduction, the management board should also consider this as a focus in the long-term development strategy. The author suggests the following policies:

Based on the company's risk appetite, the management board should effectively allocate capital among the following segments: proprietary trading, brokerage, underwriting and margin lending... to maximize risk-adjusted profits.

Avoid “chasing returns” during overheated market phases without adequate risk management frameworks, which could trigger cascading failures.

Accumulate reserve funds and maintain a high capital adequacy ratio to be ready to respond to sudden fluctuations, while maintaining the trust of shareholders, customers and management agencies.

5.2.7. GDP Growth (GDP)

In the research model, the GDP variable shows a positive impact on bankruptcy risk and a negative impact on the profitability of securities companies - a result that is somewhat contrary to traditional theoretical expectations. This finding suggests that management should reassess the influence of macroeconomic

factors on the industry and adopt appropriate strategies. Specifically, the author has some suggestions as follows:

Maintain rigorous risk management principles even during favorable periods, rather than loosening financial discipline in pursuit of short-term profits.

Closely monitor macroeconomic indicators such as economic cycles, money supply, and credit growth to formulate flexible and context-appropriate management strategies.

Strengthen risk provisions and establish profit stabilization reserves to mitigate unexpected shocks during periods of economic adjustment.

5.3. Limitations and suggestions for further studies

5.3.1. Limitations

Although the topic has met most of the research objectives, the author still found many limitations that affect the research results, including:

Further research on the relationship of macro factors: because the research objective is mainly to assess the importance of the competencies and qualifications of the executive board of companies, macro variables have not been considered for inclusion in the model. However, in the case of research that needs to delve into the bidirectional relationship between risk and profit, it is still necessary to delve deeper into macro fluctuations such as inflation rate, interest rates, etc.

Expanding research sample size: this study only uses a sample of 22 securities companies in the period from 2011 to 2024. Although in reality, about 80 securities companies are operating in Vietnam, according to the State Securities Commission, the research does not fully reflect the reality of all securities companies.

Although the study was conducted to evaluate the performance of companies, there are still many variables that have not reached statistical significance, like EDR.

5.3.2. Suggestions for further studies

Based on the above limitations, the author suggests the following future research directions:

Expand the sample size to include securities companies in the list of securities companies licensed by the State Securities Commission.

Further study the impact of macroeconomic factors such as bank interest rates, and inflation rate on the bidirectional relationship between risk and profit.

Further evaluate the qualitative factors that influence the relationship between bankruptcy risk and profitability, such as corporate culture, corporate governance, management quality...

CHAPTER 5 SUMMARY

In Chapter 5, the author has summarized the research results and made comments and assessments on the bidirectional relationship between bankruptcy risk and profitability of securities companies. Based on that, the author has made some policy recommendations for the management of securities companies to increase the operational efficiency of these companies, including promoting investment, building an advanced risk management system, promoting and developing high-quality human resources, and continuously improving infrastructure.

In addition, Chapter 5 also points out the limitations that the current research is facing. At the same time, comments and recommendations are also given for future studies to continue pursuing this topic.

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APPENDIX

Appendix 1. List of 22 Vietnamese securities companies in the research model

No.	Ticker	Securities Company Name
1	SSI	SSI Securities Corporation
2	HCM	Ho Chi Minh City Securities Corporation
3	MBS	MB Securities Joint Stock Company
4	VND	VNDIRECT Securities Corporation
5	VDS	Viet Dragon Securities Corporation
6	VIX	VIX Securities Joint Stock Company
7	SHS	Saigon - Hanoi Securities Joint Stock Company
8	AGR	Agribank Securities Corporation
9	BVS	Bao Viet Securities Joint Stock Company
10	BSI	BIDV Securities Joint Stock Company
11	CTS	Vietnam Bank For Industry And Trade Securities Joint Stock Company
12	TVS	Thien Viet Securities Joint Stock Company
13	ORS	Tien Phong Securities Corporation
14	APG	APG Securities Joint Stock Company
15	IVS	Vietnam Investment Securities Joint Stock Company
16	TVB	Tri Viet Securities Joint Stock Company
17	SBS	SBS Securities Joint Stock Company
18	APS	Asia-Pacific Securities Joint Stock Company
19	VIG	Viet Nam Financial Investment Securities Corporation
20	HAC	Hai Phong Securities Joint Stock Company
21	WSS	Wall Street Securities Joint Stock Company
22	HBS	Hoa Binh Securities Joint Stock Company

Appendix 2: Research data table

Ticker	Year	ZSCORE	ROA	SIZE	EDR	AT	RMC	GDP
SSI	2011	4.43	1.08%	13	3.82	0.11	2.43%	6.41%
SSI	2012	2.61	6.43%	13	1.95	0.12	9.80%	5.50%
SSI	2013	2.84	5.32%	13	2.24	0.09	10.88%	5.55%
SSI	2014	2.87	8.96%	13	1.84	0.19	11.41%	6.42%
SSI	2015	2.46	7.97%	13	1.24	0.15	15.50%	6.99%
SSI	2016	2.99	7.22%	13	1.30	0.19	10.62%	6.69%
SSI	2017	2.03	7.09%	13	0.85	0.18	7.72%	6.81%
SSI	2018	1.54	6.12%	13	0.62	0.17	5.04%	7.08%
SSI	2019	1.44	3.57%	13	0.53	0.13	2.12%	7.02%
SSI	2020	1.19	4.00%	14	0.38	0.14	0.71%	2.91%
SSI	2021	1.12	6.23%	14	0.39	0.17	2.01%	2.58%
SSI	2022	1.98	3.30%	14	0.75	0.12	4.17%	8.02%
SSI	2023	1.30	3.78%	14	0.51	0.12	3.08%	5.05%
SSI	2024	1.59	3.99%	14	0.57	0.12	4.03%	7.09%
HCM	2011	5.03	7.55%	12	3.39	0.19	23.40%	6.41%
HCM	2012	3.34	8.46%	13	2.09	0.19	15.36%	5.50%
HCM	2013	4.53	8.92%	12	2.55	0.20	19.28%	5.55%
HCM	2014	3.31	10.77%	13	1.57	0.24	15.97%	6.42%
HCM	2015	4.65	6.13%	12	2.77	0.17	18.36%	6.99%
HCM	2016	5.76	9.06%	13	1.95	0.24	15.21%	6.69%

HCM	2017	2.25	10.76%	13	0.72	0.30	5.93%	6.81%
HCM	2018	3.32	11.32%	13	1.39	0.39	9.14%	7.08%
HCM	2019	3.43	6.79%	13	1.35	0.24	8.75%	7.02%
HCM	2020	1.78	5.31%	13	0.55	0.23	2.86%	2.91%
HCM	2021	1.25	6.22%	13	0.43	0.24	2.27%	2.58%
HCM	2022	2.40	4.28%	13	1.04	0.20	5.57%	8.02%
HCM	2023	1.99	4.04%	13	0.87	0.17	4.89%	5.05%
HCM	2024	1.46	4.22%	13	0.50	0.17	2.66%	7.09%
MBS	2011	(0.43)	-12.39%	12	0.36	0.14	0.00%	6.41%
MBS	2012	0.13	0.41%	12	0.41	0.14	0.00%	5.50%
MBS	2013	0.73	0.48%	12	0.86	0.06	1.00%	5.55%
MBS	2014	0.55	2.55%	12	0.74	0.14	0.00%	6.42%
MBS	2015	0.78	0.29%	12	1.06	0.14	0.00%	6.99%
MBS	2016	0.91	0.55%	13	0.60	0.17	0.00%	6.69%
MBS	2017	1.35	0.62%	13	0.43	0.22	0.38%	6.81%
MBS	2018	2.01	4.41%	13	0.63	0.26	3.80%	7.08%
MBS	2019	1.80	5.39%	13	0.54	0.22	3.73%	7.02%
MBS	2020	1.50	4.57%	13	0.42	0.19	1.96%	2.91%
MBS	2021	1.41	6.50%	13	0.45	0.25	2.96%	2.58%
MBS	2022	2.04	4.74%	13	0.73	0.18	4.11%	8.02%
MBS	2023	1.53	4.50%	13	0.49	0.14	3.31%	5.05%
MBS	2024	1.51	3.97%	13	0.45	0.17	2.59%	7.09%
VND	2011	2.64	-9.14%	12	2.31	0.12	-33.98%	6.41%
VND	2012	2.87	5.04%	12	2.08	0.15	14.09%	5.50%
VND	2013	2.41	6.95%	12	1.60	0.15	16.84%	5.55%
VND	2014	2.53	6.44%	12	1.66	0.18	13.35%	6.42%
VND	2015	1.40	4.78%	13	0.71	0.13	6.50%	6.99%
VND	2016	1.74	3.79%	13	0.60	0.15	3.24%	6.69%
VND	2017	1.23	7.26%	13	0.45	0.18	2.98%	6.81%
VND	2018	1.14	4.01%	13	0.40	0.17	2.07%	7.08%
VND	2019	1.37	3.44%	13	0.39	0.14	2.00%	7.02%
VND	2020	1.51	5.20%	13	0.34	0.16	1.20%	2.91%
VND	2021	1.13	9.15%	14	0.36	0.23	2.89%	2.58%
VND	2022	1.69	3.21%	14	0.60	0.18	2.60%	8.02%
VND	2023	1.51	5.02%	14	0.65	0.16	4.58%	5.05%
VND	2024	1.61	3.99%	14	0.80	0.12	6.03%	7.09%
VDS	2011	0.17	-12.67%	12	0.41	0.16	-5.54%	6.41%
VDS	2012	0.66	-3.65%	12	0.65	0.13	-3.57%	5.50%
VDS	2013	0.71	0.24%	12	0.53	0.15	0.17%	5.55%
VDS	2014	1.14	11.79%	12	0.45	0.26	4.12%	6.42%
VDS	2015	2.32	1.98%	12	2.38	0.11	8.25%	6.99%
VDS	2016	1.59	3.78%	12	0.92	0.18	4.74%	6.69%
VDS	2017	1.99	6.48%	12	1.30	0.21	9.75%	6.81%
VDS	2018	1.91	3.64%	12	1.35	0.22	5.17%	7.08%
VDS	2019	1.54	1.64%	12	0.83	0.16	2.05%	7.02%
VDS	2020	2.55	6.17%	12	0.83	0.19	6.56%	2.91%
VDS	2021	1.69	12.93%	13	0.69	0.31	6.75%	2.58%
VDS	2022	1.81	-2.77%	13	0.96	0.20	-3.19%	8.02%
VDS	2023	1.90	6.90%	13	0.83	0.16	7.69%	5.05%
VDS	2024	1.67	4.97%	13	0.78	0.17	5.15%	7.09%
VIX	2011	23.04	1.79%	12	48.97	0.15	10.70%	6.41%
VIX	2012	16.68	-16.21%	11	31.02	0.13	-107.73%	5.50%
VIX	2013	22.19	14.31%	12	42.22	0.18	74.97%	5.55%
VIX	2014	7.24	21.11%	12	8.50	0.37	44.18%	6.42%
VIX	2015	11.03	11.76%	12	19.61	0.26	36.17%	6.99%
VIX	2016	3.14	5.63%	12	2.71	0.20	13.74%	6.69%
VIX	2017	3.57	6.48%	12	2.32	0.32	12.96%	6.81%
VIX	2018	3.46	15.18%	12	2.59	0.32	26.73%	7.08%
VIX	2019	7.85	7.06%	12	4.11	0.27	18.39%	7.02%
VIX	2020	9.58	16.92%	12	4.78	0.37	35.25%	2.91%
VIX	2021	4.86	21.50%	13	3.89	0.46	33.60%	2.58%

VIX	2022	26.84	4.85%	13	27.59	0.18	24.38%	8.02%
VIX	2023	22.97	11.21%	13	34.31	0.19	56.35%	5.05%
VIX	2024	4.68	4.62%	13	4.50	0.13	24.23%	7.09%
SHS	2011	0.32	-23.75%	12	1.72	0.14	-58.98%	6.41%
SHS	2012	1.22	2.31%	12	1.08	0.27	2.08%	5.50%
SHS	2013	0.59	0.68%	12	0.69	0.08	1.34%	5.55%
SHS	2014	0.67	4.79%	13	0.40	0.16	2.43%	6.42%
SHS	2015	1.33	3.91%	12	0.56	0.17	4.57%	6.99%
SHS	2016	1.85	2.80%	13	0.51	0.18	3.17%	6.69%
SHS	2017	1.82	9.92%	13	0.55	0.29	5.73%	6.81%
SHS	2018	2.05	7.95%	13	0.62	0.28	7.94%	7.08%
SHS	2019	2.28	4.65%	13	0.80	0.20	4.56%	7.02%
SHS	2020	3.18	11.60%	13	0.87	0.28	10.99%	2.91%
SHS	2021	2.51	15.65%	13	1.25	0.32	16.27%	2.58%
SHS	2022	8.44	1.49%	13	6.45	0.14	7.91%	8.02%
SHS	2023	7.59	5.00%	13	8.60	0.13	30.74%	5.05%
SHS	2024	4.51	7.97%	13	3.98	0.16	32.57%	7.09%
AGR	2011	0.59	1.05%	13	0.60	0.13	0.15%	6.41%
AGR	2012	0.54	1.09%	13	0.62	0.15	0.37%	5.50%
AGR	2013	0.80	0.41%	13	1.39	0.08	0.70%	5.55%
AGR	2014	2.03	0.79%	12	4.80	0.09	3.97%	6.42%
AGR	2015	21.66	(0.08)	12	90.06	0.07	-59.87%	6.99%
AGR	2016	73.08	(0.22)	12	136.86	0.07	-220.07%	6.69%
AGR	2017	49.51	3.798%	12	137.66	0.11	25.51%	6.81%
AGR	2018	14.09	3.67%	12	40.09	0.10	33.50%	7.08%
AGR	2019	12.64	3.39%	12	13.62	0.10	24.30%	7.02%
AGR	2020	15.73	4.40%	12	11.05	0.10	28.15%	2.91%
AGR	2021	6.55	15.53%	12	8.50	0.16	77.37%	2.58%
AGR	2022	9.41	5.28%	12	8.11	0.13	29.82%	8.02%
AGR	2023	5.62	5.05%	12	4.41	0.12	24.87%	5.05%
AGR	2024	3.44	4.14%	13	2.37	0.13	14.71%	7.09%
BVS	2011	2.96	-6.70%	12	3.39	0.13	-19.88%	6.41%
BVS	2012	2.84	5.35%	12	2.75	0.14	14.80%	5.50%
BVS	2013	2.33	5.16%	12	2.09	0.12	16.61%	5.55%
BVS	2014	2.81	7.16%	12	2.47	0.17	18.67%	6.42%
BVS	2015	5.67	6.54%	12	6.68	0.15	25.87%	6.99%
BVS	2016	4.78	5.77%	12	5.04	0.16	17.18%	6.69%
BVS	2017	3.08	6.04%	12	3.39	0.23	11.64%	6.81%
BVS	2018	1.74	4.04%	12	1.37	0.20	6.38%	7.08%
BVS	2019	4.22	5.09%	12	3.06	0.20	16.02%	7.02%
BVS	2020	2.25	4.56%	13	1.22	0.20	6.14%	2.91%
BVS	2021	1.41	6.20%	13	0.60	0.25	3.37%	2.58%
BVS	2022	2.31	3.07%	13	1.30	0.18	5.02%	8.02%
BVS	2023	1.41	3.97%	13	0.63	0.18	2.90%	5.05%
BVS	2024	1.61	3.43%	13	0.79	0.17	3.09%	7.09%
BSI	2011	0.18	-8.94%	12	0.26	0.08	-3.60%	6.41%
BSI	2012	1.57	0.95%	12	1.08	0.10	2.10%	5.50%
BSI	2013	1.28	1.06%	12	0.82	0.15	1.32%	5.55%
BSI	2014	1.54	4.45%	12	0.71	0.31	2.35%	6.42%
BSI	2015	1.62	5.12%	12	0.70	0.25	3.31%	6.99%
BSI	2016	4.00	6.75%	12	2.14	0.23	12.39%	6.69%
BSI	2017	2.03	9.21%	12	1.06	0.30	7.54%	6.81%
BSI	2018	6.14	9.52%	12	4.55	0.44	13.33%	7.08%
BSI	2019	2.76	5.47%	12	1.37	0.29	5.91%	7.02%
BSI	2020	3.32	4.80%	12	1.04	0.34	3.49%	2.91%
BSI	2021	1.31	8.00%	13	0.40	0.30	1.78%	2.58%
BSI	2022	6.78	1.94%	13	3.57	0.19	6.12%	8.02%
BSI	2023	2.84	5.87%	13	1.29	0.18	10.09%	5.05%
BSI	2024	2.48	4.43%	13	0.96	0.15	6.31%	7.09%
CTS	2011	1.57	5.07%	12	1.42	0.15	8.44%	6.41%
CTS	2012	3.47	5.64%	12	4.39	0.14	19.62%	5.50%

CTS	2013	3.60	6.00%	12	4.93	0.14	22.53%	5.55%
CTS	2014	2.90	5.98%	12	4.34	0.16	7.09%	6.42%
CTS	2015	12.07	7.03%	12	19.10	0.21	28.01%	6.99%
CTS	2016	3.71	6.74%	12	3.05	0.19	18.93%	6.69%
CTS	2017	2.29	6.74%	12	1.75	0.18	14.29%	6.81%
CTS	2018	1.95	6.82%	12	1.04	0.30	6.54%	7.08%
CTS	2019	1.93	4.27%	12	0.88	0.19	4.76%	7.02%
CTS	2020	1.74	3.62%	13	0.51	0.17	2.31%	2.91%
CTS	2021	1.17	7.05%	13	0.37	0.19	2.67%	2.58%
CTS	2022	1.79	1.21%	13	0.58	0.15	1.09%	8.02%
CTS	2023	1.22	2.78%	13	0.32	0.17	0.91%	5.05%
CTS	2024	1.27	2.69%	13	0.35	0.12	1.45%	7.09%
TVS	2011	17.05	6.01%	12	9.23	0.51	9.22%	6.41%
TVS	2012	1.98	7.39%	12	1.25	0.29	7.68%	5.50%
TVS	2013	3.26	9.97%	12	2.57	0.22	15.34%	5.55%
TVS	2014	3.07	11.19%	12	2.14	0.25	20.08%	6.42%
TVS	2015	7.38	12.69%	12	7.13	0.29	31.56%	6.99%
TVS	2016	1.53	6.64%	12	0.81	0.17	7.70%	6.69%
TVS	2017	2.21	11.31%	12	0.86	0.28	8.57%	6.81%
TVS	2018	1.43	6.54%	12	0.48	0.23	3.59%	7.08%
TVS	2019	1.36	3.89%	12	0.49	0.15	2.77%	7.02%
TVS	2020	1.98	6.75%	13	0.50	0.18	4.19%	2.91%
TVS	2021	1.22	9.32%	13	0.32	0.19	2.93%	2.58%
TVS	2022	0.35	0.53%	13	0.23	0.12	-0.09%	8.02%
TVS	2023	0.46	2.19%	13	0.18	0.12	0.06%	5.05%
TVS	2024	1.20	2.69%	13	0.46	0.11	1.84%	7.09%
ORS	2011	0.07	-3.51%	12	0.17	0.14	-0.28%	6.41%
ORS	2012	0.13	0.04%	12	0.39	0.02	0.07%	5.50%
ORS	2013	(1.06)	-18.76%	12	0.17	0.01	-9.34%	5.55%
ORS	2014	(0.31)	0.12%	12	0.20	0.02	0.09%	6.42%
ORS	2015	(0.11)	0.79%	12	0.22	0.12	0.19%	6.99%
ORS	2016	(0.48)	-2.92%	12	0.19	0.03	-2.28%	6.69%
ORS	2017	(0.39)	-0.66%	12	0.18	0.04	-0.35%	6.81%
ORS	2018	3.07	-4.06%	11	3.86	0.08	-28.66%	7.08%
ORS	2019	10.09	27.23%	12	4.81	0.87	19.87%	7.02%
ORS	2020	2.65	7.07%	12	0.78	0.31	8.80%	2.91%
ORS	2021	2.34	6.05%	13	0.84	0.39	6.20%	2.58%
ORS	2022	2.29	2.37%	13	0.52	0.47	1.34%	8.02%
ORS	2023	2.44	3.35%	13	0.58	0.41	2.34%	5.05%
ORS	2024	1.61	4.18%	13	0.54	0.23	3.92%	7.09%
APG	2011	25.12	-31.25%	11	46.22	0.10	-263.56%	6.41%
APG	2012	63.58	0.41%	11	124.04	0.08	4.91%	5.50%
APG	2013	1.80	3.83%	11	2.29	0.10	18.15%	5.55%
APG	2014	3.00	5.55%	11	4.80	0.05	79.36%	6.42%
APG	2015	6.10	16.81%	11	7.98	0.22	57.78%	6.99%
APG	2016	5.48	0.08%	11	12.20	0.03	1.97%	6.69%
APG	2017	33.74	5.62%	11	59.51	0.11	46.76%	6.81%
APG	2018	26.85	4.91%	11	38.11	0.11	40.55%	7.08%
APG	2019	32.90	8.92%	12	31.91	0.49	6.27%	7.02%
APG	2020	25.27	6.30%	12	25.85	0.09	25.25%	2.91%
APG	2021	5.89	33.81%	12	6.12	0.48	42.35%	2.58%
APG	2022	24.26	-13.70%	12	28.19	0.13	-75.37%	8.02%
APG	2023	15.65	8.17%	12	27.38	0.15	40.15%	5.05%
APG	2024	6.01	-5.80%	12	8.91	0.05	-71.59%	7.09%
IVS	2011	6.09	0.04%	11	4.79	0.16	0.12%	6.41%
IVS	2012	4.52	1.05%	11	3.63	0.15	2.61%	5.50%
IVS	2013	3.37	0.53%	11	2.64	0.12	1.55%	5.55%
IVS	2014	3.49	3.92%	11	2.58	0.15	9.15%	6.42%
IVS	2015	31.29	2.60%	11	32.50	0.14	12.12%	6.99%
IVS	2016	53.30	0.04%	12	86.66	0.09	0.35%	6.69%
IVS	2017	57.10	0.10%	12	71.14	0.07	1.13%	6.81%

IVS	2018	45.76	0.19%	12	72.16	0.07	2.16%	7.08%
IVS	2019	89.65	-11.41%	12	129.90	0.05	-149.85%	7.02%
IVS	2020	96.94	1.39%	12	116.51	0.04	19.16%	2.91%
IVS	2021	3.59	4.59%	12	5.41	0.11	25.24%	2.58%
IVS	2022	7.06	2.09%	12	8.77	0.09	18.59%	8.02%
IVS	2023	49.33	3.43%	12	71.27	0.10	30.21%	5.05%
IVS	2024	4.74	2.05%	12	5.68	0.09	16.05%	7.09%
TVB	2011	3.99	-7.74%	11	3.06	0.12	-31.99%	6.41%
TVB	2012	14.65	-3.23%	11	12.12	0.09	-27.28%	5.50%
TVB	2013	10.19	1.01%	11	7.93	0.14	4.90%	5.55%
TVB	2014	3.57	8.28%	11	2.44	0.11	31.86%	6.42%
TVB	2015	4.12	0.54%	11	2.17	0.11	2.83%	6.99%
TVB	2016	3.95	14.26%	11	1.69	0.25	30.18%	6.69%
TVB	2017	8.59	7.32%	11	5.37	0.14	35.08%	6.81%
TVB	2018	3.76	8.83%	11	2.45	0.17	23.93%	7.08%
TVB	2019	3.41	3.86%	12	2.61	0.12	17.69%	7.02%
TVB	2020	3.69	8.44%	12	1.68	0.18	23.90%	2.91%
TVB	2021	3.51	21.60%	12	3.13	0.31	44.14%	2.58%
TVB	2022	4.13	-22.67%	12	6.66	0.11	-153.27%	8.02%
TVB	2023	30.76	6.30%	12	44.12	0.14	42.32%	5.05%
TVB	2024	20.70	11.53%	12	34.31	0.17	63.89%	7.09%
SBS	2011	(2.63)	-27.68%	12	0.00	0.16	0.12%	6.41%
SBS	2012	(1.42)	-6.80%	12	(0.09)	0.11	1.94%	5.50%
SBS	2013	0.22	42.77%	12	0.70	0.11	42.23%	5.55%
SBS	2014	(2.44)	2.33%	12	0.68	0.11	3.86%	6.42%
SBS	2015	(3.24)	1.27%	12	1.23	0.14	2.38%	6.99%
SBS	2016	(2.74)	-2.51%	12	1.24	0.17	-4.05%	6.69%
SBS	2017	(2.40)	0.13%	12	0.80	0.36	0.06%	6.81%
SBS	2018	(2.87)	0.16%	12	0.86	0.20	0.15%	7.08%
SBS	2019	(1.48)	0.04%	12	0.90	0.15	0.06%	7.02%
SBS	2020	0.36	0.23%	12	0.58	0.18	0.15%	2.91%
SBS	2021	(0.81)	1.16%	12	0.40	0.34	0.25%	2.58%
SBS	2022	(0.01)	-10.24%	12	1.19	0.20	-13.71%	8.02%
SBS	2023	(0.36)	0.08%	12	1.10	0.23	0.09%	5.05%
SBS	2024	(1.45)	-12.18%	12	1.17	0.21	-14.64%	7.09%
APS	2011	1.22	-10.98%	12	3.03	0.11	-55.08%	6.41%
APS	2012	2.22	3.55%	12	5.31	0.08	30.22%	5.50%
APS	2013	1.98	2.95%	12	4.86	0.06	30.23%	5.55%
APS	2014	2.02	4.26%	12	3.94	0.11	23.59%	6.42%
APS	2015	7.54	1.60%	12	25.48	0.10	14.15%	6.99%
APS	2016	15.40	0.72%	12	42.11	0.13	5.01%	6.69%
APS	2017	13.81	2.71%	12	24.76	0.31	7.78%	6.81%
APS	2018	14.56	0.50%	12	24.38	0.43	1.04%	7.08%
APS	2019	12.75	-9.15%	12	23.55	0.07	-117.05%	7.02%
APS	2020	72.79	14.71%	12	36.89	0.41	32.78%	2.91%
APS	2021	9.05	55.61%	12	6.97	0.74	60.08%	2.58%
APS	2022	52.92	-34.71%	12	69.54	0.33	-97.43%	8.02%
APS	2023	34.65	-20.24%	12	36.43	0.49	-35.80%	5.05%
APS	2024	43.01	2.69%	12	66.40	0.29	8.10%	7.09%
VIG	2011	0.70	0.00%	12	3.09	0.14	0.00%	6.41%
VIG	2012	0.64	0.00%	11	5.57	0.06	0.00%	5.50%
VIG	2013	1.70	0.00%	11	6.95	0.05	0.00%	5.55%
VIG	2014	1.62	3.78%	11	5.93	0.05	50.16%	6.42%
VIG	2015	1.99	2.47%	11	9.15	0.03	56.29%	6.99%
VIG	2016	3.54	-9.99%	11	11.40	0.03	-271.95%	6.69%
VIG	2017	2.18	0.50%	11	12.10	0.04	9.41%	6.81%
VIG	2018	1.80	0.70%	11	15.17	0.02	24.20%	7.08%
VIG	2019	4.08	-10.78%	11	10.96	0.03	-321.25%	7.02%
VIG	2020	12.37	-1.65%	11	8.82	0.03	-42.74%	2.91%
VIG	2021	4.78	0.67%	11	8.36	0.05	9.39%	2.58%
VIG	2022	23.04	0.99%	12	24.81	0.27	3.30%	8.02%

VIG	2023	17.00	9.34%	12	28.50	0.12	38.84%	5.05%
VIG	2024	13.69	5.77%	12	21.55	0.12	25.91%	7.09%
HAC	2011	1.66	-14.21%	11	4.91	0.16	-46.57%	6.41%
HAC	2012	1.06	-29.43%	12	4.48	0.19	-76.01%	5.50%
HAC	2013	1.55	3.60%	12	3.93	0.09	21.51%	5.55%
HAC	2014	1.77	4.12%	12	3.97	0.11	22.12%	6.42%
HAC	2015	3.24	4.84%	12	5.84	0.12	25.14%	6.99%
HAC	2016	14.40	5.30%	12	39.12	0.10	43.32%	6.69%
HAC	2017	13.14	1.63%	11	30.24	0.11	11.20%	6.81%
HAC	2018	13.59	-2.57%	11	28.90	0.14	-15.59%	7.08%
HAC	2019	23.50	1.81%	11	36.11	0.19	8.38%	7.02%
HAC	2020	19.23	6.98%	12	21.48	0.19	32.53%	2.91%
HAC	2021	6.90	10.86%	12	9.77	0.25	34.25%	2.58%
HAC	2022	12.14	-26.00%	11	12.51	0.20	-106.29%	8.02%
HAC	2023	30.78	12.88%	11	26.78	0.40	28.85%	5.05%
HAC	2024	20.30	1.00%	11	21.92	0.23	3.86%	7.09%
WSS	2011	14.94	0.24%	12	41.18	0.08	2.02%	6.41%
WSS	2012	10.18	-0.23%	12	29.30	0.06	-1.92%	5.50%
WSS	2013	25.48	0.15%	12	82.14	0.04	2.43%	5.55%
WSS	2014	14.53	1.73%	12	36.08	0.12	11.48%	6.42%
WSS	2015	38.22	1.31%	12	121.44	0.08	14.56%	6.99%
WSS	2016	41.50	0.93%	12	132.73	0.07	10.97%	6.69%
WSS	2017	4.37	2.68%	12	11.78	0.11	19.54%	6.81%
WSS	2018	3.81	7.21%	12	9.44	0.15	37.66%	7.08%
WSS	2019	4.23	-13.15%	12	10.55	0.04	-229.70%	7.02%
WSS	2020	112.56	1.97%	12	138.51	0.06	28.87%	2.91%
WSS	2021	38.52	1.68%	12	102.35	0.07	22.29%	2.58%
WSS	2022	33.27	-3.52%	12	74.68	0.07	-48.78%	8.02%
WSS	2023	49.94	0.32%	12	138.57	0.08	3.57%	5.05%
WSS	2024	33.68	-6.24%	12	103.64	0.04	-139.55%	7.09%
HBS	2011	7.25	0.32%	12	20.85	0.11	1.62%	6.41%
HBS	2012	12.01	0.06%	12	41.26	0.09	0.42%	5.50%
HBS	2013	17.33	0.08%	12	70.21	0.06	0.88%	5.55%
HBS	2014	5.06	0.54%	12	20.85	0.03	6.15%	6.42%
HBS	2015	18.04	0.34%	12	113.77	0.03	5.18%	6.99%
HBS	2016	52.72	0.52%	12	291.68	0.03	6.97%	6.69%
HBS	2017	2.58	0.72%	12	14.44	0.02	12.87%	6.81%
HBS	2018	49.35	0.66%	12	342.33	0.03	12.47%	7.08%
HBS	2019	77.37	1.92%	12	256.86	0.04	27.39%	7.02%
HBS	2020	86.35	1.15%	12	143.00	0.03	17.93%	2.91%
HBS	2021	27.66	2.04%	12	88.31	0.05	25.82%	2.58%
HBS	2022	52.80	1.20%	12	111.80	0.07	14.28%	8.02%
HBS	2023	39.88	4.28%	12	98.75	0.09	39.44%	5.05%
HBS	2024	25.94	2.47%	12	59.89	0.07	28.24%	7.09%

Appendix 3: Regression results

Descriptive statistics of variables:

```
. sum
```

Variable	Obs	Mean	Std. dev.	Min	Max
Ticker	0				
Year	308	2017.5	4.037689	2011	2024
ZSCORE	308	9.767946	17.08503	-3.244711	112.5642
ROA	308	.0300073	.0879648	-.3470639	.5560888
SIZE	308	12.15758	.639503	10.85512	13.86633
EDR	308	18.23167	40.97861	-.0880373	342.3282
AT	308	.1696984	.1079906	.0148252	.8683537
RMC	308	.0210409	.4443937	-3.212496	.7936418
GDP	308	.0600857	.0152682	.0258	.0802

Analysis of the correlation matrix between variables:

```
. pwcorr
```

(Ticker ignored because string variable)

	Year	ZSCORE	ROA	SIZE	EDR	AT	RMC
Year	1.0000						
ZSCORE	0.1521	1.0000					
ROA	0.1519	-0.1320	1.0000				
SIZE	0.3806	-0.3169	0.2036	1.0000			
EDR	0.0645	0.8131	-0.1523	-0.3060	1.0000		
AT	0.1922	-0.1439	0.4396	0.1898	-0.2752	1.0000	
RMC	0.0368	-0.0870	0.6382	0.1109	-0.0772	0.1840	1.0000
GDP	-0.1157	-0.0560	-0.2742	-0.0917	0.0283	-0.1471	-0.1676
	GDP						
GDP	1.0000						

Summary of regression results using FEM, REM:

Bankruptcy risk model:

Regression results using FEM model:

```
. xtreg ZSCORE ROA SIZE EDR AT RMC GDP, fe
note: GDP omitted because of collinearity.
```

```
Fixed-effects (within) regression      Number of obs   =       308
Group variable: Year                  Number of groups =       14

R-squared:                            Obs per group:
    Within = 0.7014                      min =          22
    Between = 0.3271                     avg =         22.0
    Overall = 0.6716                      max =          22

F(5,289) = 135.75
corr(u_i, Xb) = 0.0040                  Prob > F = 0.0000
```

ZSCORE	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ROA	-3.165581	9.681537	-0.33	0.744	-22.22084	15.88968
SIZE	-3.981499	.9732073	-4.09	0.000	-5.896972	-2.066026
EDR	.3247921	.014583	22.27	0.000	.2960898	.3534944
AT	14.44467	6.013999	2.40	0.017	2.607882	26.28147
RMC	-.5413343	1.62993	-0.33	0.740	-3.749373	2.666704
GDP	(omitted)					
_cons	49.907	11.97765	4.17	0.000	26.33251	73.48149
sigma_u	4.0911043					
sigma_e	9.2331765					
rho	.16410776	(fraction of variance due to u_i)				

F test that all u_i=0: F(13, 289) = 3.95 Prob > F = 0.0000

Regression results using REM model:

```
. xtreg ZSCORE ROA SIZE EDR AT RMC GDP, re
```

```
Random-effects GLS regression      Number of obs   =       308
Group variable: Year              Number of groups =       14

R-squared:                        Obs per group:
    Within = 0.7008                      min =          22
    Between = 0.4699                     avg =         22.0
    Overall = 0.6817                      max =          22

Wald chi2(6) = 683.38
corr(u_i, X) = 0 (assumed)          Prob > chi2 = 0.0000
```

ZSCORE	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
ROA	-5.910733	9.413051	-0.63	0.530	-24.35997	12.53851
SIZE	-3.335246	.9437189	-3.53	0.000	-5.184901	-1.485591
EDR	.3297101	.0144059	22.89	0.000	.3014751	.3579451
AT	15.64948	5.913886	2.65	0.008	4.058481	27.24049
RMC	-.5728059	1.61253	-0.36	0.722	-3.733306	2.587694
GDP	-96.42117	61.73189	-1.56	0.118	-217.4135	24.57111
_cons	47.63257	12.29943	3.87	0.000	23.52613	71.73902
sigma_u	2.8315769					
sigma_e	9.2331765					
rho	.0859642	(fraction of variance due to u_i)				

Profitability model:

Regression results using FEM model:

```
. xtreg ROA ZSCORE SIZE EDR AT RMC GDP, fe
note: GDP omitted because of collinearity.
```

```
Fixed-effects (within) regression      Number of obs   =       308
Group variable: Year                  Number of groups =       14

R-squared:                            Obs per group:
    Within = 0.5008                      min =          22
    Between = 0.6886                     avg =         22.0
    Overall = 0.5206                     max =          22

corr(u_i, Xb) = 0.1858                  F(5,289)        =       57.98
                                      Prob > F          =       0.0000
```

ROA	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ZSCORE	-.0001168	.0003573	-0.33	0.744	-.00082	.0005864
SIZE	.0116584	.006042	1.93	0.055	-.0002334	.0235502
EDR	8.64e-06	.000146	0.06	0.953	-.0002787	.000296
AT	.2387582	.0341186	7.00	0.000	.1716057	.3059106
RMC	.1022543	.0078673	13.00	0.000	.0867698	.1177388
GDP	0 (omitted)					
_cons	-.1534152	.074369	-2.06	0.040	-.2997888	-.0070417
sigma_u	.02883913					
sigma_e	.05608899					
rho	.20909114	(fraction of variance due to u_i)				

F test that all u_i=0: F(13, 289) = 5.56 Prob > F = 0.0000

Regression results using REM model:

```
. xtreg ROA ZSCORE SIZE EDR AT RMC GDP, re
```

```
Random-effects GLS regression      Number of obs   =       308
Group variable: Year              Number of groups =       14

R-squared:                        Obs per group:
    Within = 0.5007                      min =          22
    Between = 0.6727                     avg =         22.0
    Overall = 0.5377                     max =          22

Wald chi2(6)                      =       311.38
Prob > chi2                        =       0.0000

corr(u_i, X) = 0 (assumed)
```

ROA	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
ZSCORE	-.0001742	.0003513	-0.50	0.620	-.0008628	.0005145
SIZE	.0110731	.005887	1.88	0.060	-.0004653	.0226115
EDR	.0000328	.0001446	0.23	0.820	-.0002506	.0003162
AT	.241574	.0336304	7.18	0.000	.1756597	.3074883
RMC	.1034808	.0077682	13.32	0.000	.0882553	.1187062
GDP	-.7946282	.5063093	-1.57	0.117	-1.786976	.1977198
_cons	-.0989377	.0801458	-1.23	0.217	-.2560207	.0581453
sigma_u	.02615371					
sigma_e	.05608899					
rho	.17859486	(fraction of variance due to u_i)				

Hausman test results:

Bankruptcy risk model:

```
. hausman fe re
```

	Coefficients			
	(b) fe	(B) re	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
ROA	-3.165581	-5.910733	2.745152	2.264206
SIZE	-3.981499	-3.335246	-.6462531	.2377546
EDR	.3247921	.3297101	-.004918	.0022659
AT	14.44467	15.64948	-1.20481	1.092767
RMC	-.5413343	-.5728059	.0314716	.2375287

b = Consistent under H0 and Ha; obtained from **xtreg**.

B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

```
chi2(5) = (b-B)'[(V_b-V_B)^(-1)](b-B)
        = 8.48
Prob > chi2 = 0.1317
(V_b-V_B is not positive definite)
```

Profitability model:

```
. hausman fe re
```

	Coefficients			
	(b) fe	(B) re	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
ZSCORE	-.0001168	-.0001742	.0000573	.0000648
SIZE	.0116584	.0110731	.0005853	.0013594
EDR	8.64e-06	.0000328	-.0000242	.0000202
AT	.2387582	.241574	-.0028159	.0057513
RMC	.1022543	.1034808	-.0012265	.0012449

b = Consistent under H0 and Ha; obtained from **xtreg**.

B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

```
chi2(5) = (b-B)'[(V_b-V_B)^(-1)](b-B)
        = 2.57
Prob > chi2 = 0.7658
```

Multicollinearity test results:

Bankruptcy risk model:

```
. vif
```

Variable	VIF	1/VIF
ROA	2.16	0.463037
RMC	1.72	0.581067
AT	1.34	0.745431
EDR	1.17	0.856446
SIZE	1.14	0.877145
GDP	1.08	0.921805
Mean VIF	1.44	

Profitability model:

```
. vif
```

Variable	VIF	1/VIF
EDR	3.23	0.309417
ZSCORE	3.15	0.317915
AT	1.17	0.856721
SIZE	1.15	0.866886
GDP	1.07	0.935498
RMC	1.07	0.935520
Mean VIF	1.81	

Autocorrelation test results:

Bankruptcy risk model:

```
. xtserial ZSCORE ROA SIZE EDR AT RMC GDP
```

Wooldridge test for autocorrelation in panel data

H0: no first-order autocorrelation

```
F( 1, 21) = 1.850
Prob > F = 0.1882
```

Profitability model:

```
. xtserial ROA ZSCORE SIZE EDR AT RMC GDP
```

Wooldridge test for autocorrelation in panel data

H0: no first-order autocorrelation

```
F( 1, 21) = 14.764
Prob > F = 0.0009
```

Heteroscedasticity test results:

ZSCORE model:

```
. xttest0
```

Breusch and Pagan Lagrangian multiplier test for random effects

$$\text{ZSCORE}[\text{Year},t] = Xb + u[\text{Year}] + e[\text{Year},t]$$

Estimated results:

	Var	SD = sqrt(Var)
ZSCORE	291.8983	17.08503
e	85.25155	9.233177
u	8.017828	2.831577

Test: $\text{Var}(u) = 0$

$$\begin{aligned} \text{chibar2}(01) &= 19.98 \\ \text{Prob} > \text{chibar2} &= 0.0000 \end{aligned}$$

ROA model:

```
. xttest0
```

Breusch and Pagan Lagrangian multiplier test for random effects

$$\text{ROA}[\text{Year},t] = Xb + u[\text{Year}] + e[\text{Year},t]$$

Estimated results:

	Var	SD = sqrt(Var)
ROA	.0077378	.0879648
e	.003146	.056089
u	.000684	.0261537

Test: $\text{Var}(u) = 0$

$$\begin{aligned} \text{chibar2}(01) &= 50.77 \\ \text{Prob} > \text{chibar2} &= 0.0000 \end{aligned}$$

S-GMM model regression results:

Bankruptcy risk model:

```
. xtabond2 ZSCORE L.ZSCORE ROA SIZE EDR AT RMC GDP, gmm(l6.RMC, collapse) iv(ROA SIZE ED
> R AT GDP, eq(level)) robust
Favoring space over speed. To switch, type or click on mata: mata set matafavor speed, p
> erm.
Warning: Two-step estimated covariance matrix of moments is singular.
Using a generalized inverse to calculate robust weighting matrix for Hansen test.
Difference-in-Sargan/Hansen statistics may be negative.
```

Dynamic panel-data estimation, one-step system GMM

Group variable: ID	Number of obs	=	286
Time variable : Year	Number of groups	=	22
Number of instruments = 14	Obs per group: min	=	13
Wald chi2(7) = 895.67	avg	=	13.00
Prob > chi2 = 0.000	max	=	13

ZSCORE	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
ZSCORE						
L1.	-.2613893	.1064677	-2.46	0.014	-.4700622	-.0527165
ROA	-65.15483	24.22226	-2.69	0.007	-112.6296	-17.68008
SIZE	-3.189389	1.508423	-2.11	0.034	-6.145843	-.2329348
EDR	.3821144	.0990268	3.86	0.000	.1880254	.5762033
AT	25.55869	13.31841	1.92	0.055	-.5449123	51.6623
RMC	14.07829	5.9332	2.37	0.018	2.449429	25.70715
GDP	-157.3107	70.74967	-2.22	0.026	-295.9775	-18.64391
_cons	51.20508	23.26166	2.20	0.028	5.613067	96.79709

Instruments for first differences equation

GMM-type (missing=0, separate instruments for each period unless collapsed)
L(1/13).L6.RMC collapsed

Instruments for levels equation

Standard

ROA SIZE EDR AT GDP
_cons

GMM-type (missing=0, separate instruments for each period unless collapsed)
D.L6.RMC collapsed

Arellano-Bond test for AR(1) in first differences: z = -1.53 Pr > z = 0.126
Arellano-Bond test for AR(2) in first differences: z = -1.92 Pr > z = 0.055

Sargan test of overid. restrictions: chi2(6) = 1.78 Prob > chi2 = 0.939
(Not robust, but not weakened by many instruments.)

Hansen test of overid. restrictions: chi2(6) = 2.22 Prob > chi2 = 0.898
(Robust, but weakened by many instruments.)

Difference-in-Hansen tests of exogeneity of instrument subsets:

GMM instruments for levels

Hansen test excluding group: chi2(5) = 2.22 Prob > chi2 = 0.818
Difference (null H = exogenous): chi2(1) = 0.00 Prob > chi2 = 0.976

iv(ROA SIZE EDR AT GDP, eq(level))

Hansen test excluding group: chi2(1) = 0.76 Prob > chi2 = 0.384
Difference (null H = exogenous): chi2(5) = 1.47 Prob > chi2 = 0.917

.

Profitability model:

```
. xtabond2 ROA L1.ROA ZSCORE SIZE EDR AT RMC GDP, gmm(l7.EDR, collapse) iv(ZSCORE SIZE AT
> RMC GDP, eq(level)) robust
Favoring space over speed. To switch, type or click on mata: mata set matafavor speed, p
> erm.
```

Warning: Two-step estimated covariance matrix of moments is singular.

Using a generalized inverse to calculate robust weighting matrix for Hansen test.

Difference-in-Sargan/Hansen statistics may be negative.

Dynamic panel-data estimation, one-step system GMM

Group variable: ID	Number of obs	=	286
Time variable : Year	Number of groups	=	22
Number of instruments = 13	Obs per group: min	=	13
Wald chi2(7) = 611.12	avg	=	13.00
Prob > chi2 = 0.000	max	=	13

	ROA	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
ROA							
L1.		-.1410351	.0937662	-1.50	0.133	-.3248135	.0427433
ZSCORE		-.0015568	.0008521	-1.83	0.068	-.0032269	.0001132
SIZE		.0115962	.0060218	1.93	0.054	-.0002063	.0233986
EDR		.0006088	.0004302	1.42	0.157	-.0002344	.0014521
AT		.2939986	.0439273	6.69	0.000	.2079026	.3800946
RMC		.0988037	.0295238	3.35	0.001	.0409381	.1566692
GDP		-.6710344	.2964148	-2.26	0.024	-1.251997	-.090072
_cons		-.1093107	.0805721	-1.36	0.175	-.2672291	.0486077

Instruments for first differences equation

GMM-type (missing=0, separate instruments for each period unless collapsed)

L(1/13).L7.EDR collapsed

Instruments for levels equation

Standard

ZSCORE SIZE AT RMC GDP

_cons

GMM-type (missing=0, separate instruments for each period unless collapsed)

D.L7.EDR collapsed

Arellano-Bond test for AR(1) in first differences: z = -2.03 Pr > z = 0.043

Arellano-Bond test for AR(2) in first differences: z = -1.48 Pr > z = 0.138

Sargan test of overid. restrictions: chi2(5) = 9.01 Prob > chi2 = 0.109
(Not robust, but not weakened by many instruments.)

Hansen test of overid. restrictions: chi2(5) = 3.81 Prob > chi2 = 0.577
(Robust, but weakened by many instruments.)

Difference-in-Hansen tests of exogeneity of instrument subsets:

GMM instruments for levels

Hansen test excluding group: chi2(4) = 3.77 Prob > chi2 = 0.439

Difference (null H = exogenous): chi2(1) = 0.04 Prob > chi2 = 0.832

iv(ZSCORE SIZE AT RMC GDP, eq(level))

Hansen test excluding group: chi2(0) = 0.00 Prob > chi2 = .

Difference (null H = exogenous): chi2(5) = 3.81 Prob > chi2 = 0.577

.